

A Semantic Analysis of Cantonese *Sin*

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Abstract

In this thesis, we investigate the multifaceted uses of the Cantonese particle *sin* (先). Firstly, we argue with historical evidence that the temporal *sin* derives the question focus *sin* and the scalar *sin* separately. Secondly, we argue that the formulations of temporal *sin* derives question-focus by type-shifting: while temporal *sin* arranges the sequence of two propositions, question *sin* arranges the sequence of two questions. §4 argues that semantics of scalar *sin* gives a LONG presupposition between its prejacent *p* and a contextually salient expectation *r*. It combines with various exclusive particles by predicate modification. Decomposing the scalar and the exclusive components resolve the directionality puzzle: a single *sin* can associate with various exclusive particles, which can have either directionality.

摘要

在本論文中，我們探討粵語助詞「先」(sin) 的多重用法。首先，我們以歷史證據論證：時間義「先」分別獨立衍生出疑問焦點 (question-focus) 「先」與梯級義 (scalar) 「先」。其次，我們主張時間義「先」的形式表述透過類型轉換 (type-shifting) 衍生出疑問焦點用法：時間義「先」排列兩個命題的先後順序，而疑問「先」則排列兩個問句的先後順序。第 4 節論證，梯級義「先」的語義在其前述命題 p 與語境中顯著的預期 r 之間給出一個 LONG 預設。它透過調詞修飾與各種排他助詞結合。將梯級成分與排他成分分解，可解決方向性問題：單一的「先」可以與各種排他助詞聯結，而這些排他助詞可以具有任一方向性。

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Chapter 1

Setting the stage

1.1 The three *sins* in Cantonese

Cantonese¹ *sin* is highly multifunctional. First, it can function as a temporal adverb *sin_T* ‘first’, as in (1).²

- (1) Mary lai party *sin_T*.
Mary come party SIN
‘Mary will come to the party first.’

Second, it can be used as a question-focus particle *sin_{QF}*, roughly comparable to English focus-sensitive *even* in certain contexts (Daniels and Greenberg, 2025, Iatridou and Tatevosov, 2016). In (2), interlocutor *B* treats the question of *which Anna interlocutor A is referring to* as more fundamental than the question of *whether Anna called*.

- (2) Context: there are two Annas who are friends with both A and B.

¹Throughout this thesis, ‘Cantonese’ refers to Guangzhou and Hong Kong Cantonese.

²This thesis adopts the Jyutping romanization system for Hong Kong and Guangzhou Cantonese. Tones are transcribed only for particles.

A: Anna called. [Presupposes: there is a unique Anna.]
 B: dangzan, nei gong bingo Anna $\sin_{QF}$?
 B: wait, 2.SG say which Anna SIN
 B: ‘Wait, which Anna did you even refer to?’

Third, $\sin$ has a scalar use that is functionally similar to Mandarin *cai*. Call this use $\sin_S$. Mandarin *cai* lacks the temporal and question-focus uses found in Cantonese. $\sin_S$ is more complex still, because it frequently interacts with exclusive particles. It combines with the exclusive particle *dak* ‘only’ to yield a *so-few* evaluation, as in (3). Call this use $\sin_{Few}$, a subtype of $\sin_S$.

- (3) John $\sin_{Few}$ sik dak [saam]_F-go pinggwo.
 John SIN eat only three-CL apple
 ‘John only ate three apples. (So few!)’

$\sin_S$ also combines with the exclusive particle *zikdou* ‘not until’ to yield a *so-late* evaluation, as in (4). Call this use $\sin_{Late}$.

- (4) Context: Mary should sleep at 12am, but...
 zikdou [tingwong]_F Mary $\sin_{Late}$ fangaau.
 not.until dawn Mary SIN sleep
 ‘Mary did not sleep until dawn. (So late!)’

Finally, $\sin_S$ combines with the exclusive particle *zijau* ‘only if’ to yield a *so-picky* evaluation. That is, while a lower condition p' was expected to suffice for a consequent q , a *zijau-sin_S* construction states that only a higher condition p makes q true, as in (5). Call this use $\sin_{Picky}$.

- (5) Context: Beth is B’s good friend, and Mary is B’s best friend. A: Beth is coming. Will you come?
 B: (zijau) [Mary]_F lai, ngo $\sin_{Picky}$ lai.
 B: only.if Mary come, 1.SG SIN come

B: ‘Only if Mary comes will I come. (I am being picky.)’

Examples (3), (4) and (5) reveal what we call the directionality puzzle (Sun and Ng, Forthcoming).

(6) The directionality puzzle

- a. If the prosodic focus lies to the **right** of scalar *sin* (e.g. *sin_{Few}*), *sin* says that its prejacent is much LOWER than an expectation.
- b. If the prosodic focus lies to the **left** of scalar *sin* (e.g. *sin_{Late}* and *sin_{Picky}*), *sin* says that the relevant condition is much HIGHER than an expectation.

The same puzzle arises with other Sinitic particles, including Mandarin *jiu* and its Cantonese counterpart *zau*. A common response to the directionality of Mandarin *jiu* is to posit two lexical entries, one for each direction (Liu, 2021a, ?). This thesis argues that a single *sin* is sufficient. The directionality puzzle is caused by the exclusive particles that co-occur with *sin*, not by *sin* itself.³

1.2 The historical derivation of *sin*

Before turning to the formal analysis, we review the historical development of the particle. The reason is that the formal definitions of the various uses of *sin* should be aligned with their historical derivations.⁴

We argue that *sin_T* derives *sin_S* and *sin_{QF}* separately, as in (7).

³If the analysis of scalar *sin* is on the right track, Cantonese *zau*—and therefore Mandarin *jiu*—may likewise need only one lexical entry. See Chapter 4 for discussion.

⁴Suppose a particle has uses *X*, *Y* and *Z*. Two possible pathways are given in (i).

- (i)
 - a. $X \rightarrow Y \rightarrow Z$
 - b. $X \rightarrow Y$ and $X \rightarrow Z$

- (7) a. $\sin_T \rightarrow \sin_S$
 b. $\sin_T \rightarrow \sin_{QF}$

Previous literature suggests that Guangzhou (GZ) Cantonese *sin* has undergone the diachronic change in (8) since the Qing dynasty (Cai and Huang, 2025, Li, 2019, Wang, 2025b, Zhou, 2018).⁵

- (8) $zing$ (正) $\rightarrow$ zi (至) $\rightarrow$ $\sin zi$ (先至) $\rightarrow$ $\sin_S$ (先)

In the Qing dynasty, GZ Cantonese mainly used *zing* to express the meaning of modern Mandarin *cai* and modern Cantonese $\sin_S$, including the sequential use in (9).⁶

- (9) *saanlou keikeoi jan mei suk, soeng tiu loenggai zing sihang.*
 mountain.road rocky human NEG familiar think CL good.plan ZING execute
 ‘The mountain road is rocky and we are not familiar with it. We should think
 of a good plan before we walk.’ [Early Qing GZ Cantonese; Wang 2025b ex. 6]

Zing was gradually replaced by *zi* in the early nineteenth century. Wang (2025b) suggests that *zi* is a local Cantonese form corresponding to *zoi* (再) ‘then/again’, itself a product of contact with Mandarin. (10) is a Qing-dynasty example of *zi*.

The two pathways impose different constraints on the formalization. In both cases *X* derives *Y*, so the denotation of *Y* must be based on that of *X*. On pathway (i-a), *Y* further derives *Z*, so the denotation of *Z* must depend on *Y*; on pathway (i-b), the denotation of *Z* depends directly on *X*. This is why even a formal-semantic study needs a historical sketch.

⁵*Zi*, $\sin_S$ and $\sin_S zi$ are still used as scalar particles in modern Cantonese; *zing* is not (Wang, 2025a).

⁶Early Qing Cantonese differs slightly from Modern Cantonese in phonology, so Jyutping is only an approximation for the diachronic data. The approximation does not affect the arguments of this thesis.

- (10) doidang mingmaan zi soengloeng ne-gin si gam2 laa1.
 wait tomorrow.night ZI discuss this-CL matter PRT SFP
 ‘Let’s wait until tomorrow night to talk about this matter.’

[Qing Guangzhou Cantonese; Wang 2025b ex. 14]

Omitting the truth-conditionally vacuous particles *gam2* and *laa1*, we decompose (10) into four parts: (i) a covert subject *we* (the interlocutors); (ii) a property *P*, *doidang mingmaan* ‘wait until tomorrow night’; (iii) the particle *zi*; and (iv) a property *Q*, *soengloeng ne-gin si* ‘talk about this matter’. *Zi* says that the subject realizes the two properties in the order $P > Q$. In this respect *zi* means the same as temporal *sin*, which likewise sequences two events, as in (11).

- (11) John sik maanfaan sin. (Jinhau heoi jauseoi.)
 John eat dinner first then go swim
 ‘John had dinner first. And then he went swimming.’

[Modern HK Cantonese]

In (11), event *P* is *sik maanfaan* ‘have dinner’ and *Q* is *heoi jauseoi* ‘go swimming’. *Sin* says that John realizes the two properties in the order $P > Q$, just as *zi* does. The difference is that *sin* asserts *P* and presupposes *Q*, whereas *zi* asserts **both** *P* and *Q*. When the two clauses *P sin* and *zi Q* co-occur, that difference becomes trivial: both the antecedent *P* and the consequent *Q* are asserted, so both particles simply encode $P > Q$, as in (12).

- (12) gam zau jam-co ne-wun tong sin zi syun laa1.
 then ZAU drink-PRV this-CL soup SIN ZI think SFP
 ‘You should think about this matter after drinking the soup.’

[1910s GZ Cantonese; Wang 2025b ex. 23]

Because sin_T is postverbal,⁷ and because the two properties share a subject,⁸ (12) has the linear order in (13).

$$(13) \quad \text{Subject} > P > \text{sin}_T > \text{zi} > Q$$

As a result, sin_T and zi become adjacent and undergo fusion (univerbation): two adjacent morphs lose their boundary and are reanalyzed as a single word (???). Sinzi then follows the grammaticalization pathway of zi and develops a scalar use.

⁷In Sinitic languages, adverbials are canonically preverbal. Mandarin *xian* (先), the counterpart of Cantonese sin_T , can therefore occur only preverbally, as in (i).

- (i) Yuehan {xian} chi wanfan *{xian}.
 John first eat dinner first
 ‘John ate dinner first. (And then...)’ [Mandarin]

In Hong Kong Cantonese, by contrast, sin_T is predominantly postverbal, as in (ii).

- (ii) John ??{ sin_T } sik maanfaan { sin_T }
 John first eat dinner first
 ‘John ate dinner first. (And then...)’ [HK Cantonese]

Symbesma (2013), Zhou (2018) attribute the postverbal placement to contact with Hmong-Mien and Tai-Kadai languages.

⁸We assume that a property combines with a subject to form a proposition. Univerbation of sinzi was possible only when two properties co-occurred, not two full propositions. With two propositions the linear order would be (i).

- (i) Subject₁ > P > sin_T > Subject₂ > zi > Q

Subject₂ would then intervene between sin_T and zi , blocking adjacency and hence univerbation. Modern Cantonese $\text{sin}_S \text{zi}$ (as well as sin_S and zi) can co-occur with two propositions, as in the sin_{Late} and sin_{Picky} uses. We argue that it is the property use that first gave rise to $\text{sin}_S \text{zi}$, which later spread to the other scalar uses.

(14) is a subject-focus use, felicitous as a response to the addressee’s view that it is not good to have distinctive features.

- (14) yigaa gong daksik sinzi hou aalmaa3!
 now say feature SINZI good SFP
 ‘Having distinctive features is considered good now.’
 [1920s GZ Cantonese; Wang 2025b ex. 31]

(14) does not rank two properties in temporal sequence, so *sinzi* cannot be understood as two separate temporal adverbs. It is a single scalar particle grammaticalized from a temporal adverb.

Wang (2025b) argues that from at least the 1940s, *sin* can appear independently with the meaning of *sinzi*, whether temporal (15) or scalar (16).

- (15) ngodei tai coeng hei sin heoi sik faan aa1.
 1.PL watch CL film SIN go eat rice SFP
 ‘Let’s watch a film before we go for a meal.’
 [1965 GZ Cantonese; Wang 2025b ex. 2c]
- (16) dimgaai gam ci sin lai zek1?
 why so late SIN come SFP
 ‘Why are you coming so late?’
 [1955 GZ Cantonese; Wang 2025b ex. 1c]

In modern GZ and HK Cantonese, *sins* can be replaced by *zi* or *sinszi* because of this history, as in the *so-few* use (17), the *so-late* use (18) and the *so-picky* use (19).

- (17) John {sin_{Few}/ sin_{Few}zi/ zi} sik-dak [saam]_F-go pinggwo
 John SIN/ SINZI/ ZI eat-only three-CL apple
 ‘John just ate three apples. (So few!)’

- (18) John (zikdou) sik-jyun [maanfaan]_F {sin_{Late}/ sin_{Late}zi/ zi} heoi jauseoi.
 John not.until eat-PRV dinner SIN/ SINZI/ ZI go swim
 ‘It was not until John finished his dinner that he went swimming. (So late!)’
- (19) (zijau) [Mary]_F heoi party, John {sin_{Picky}/ sin_{Picky}zi/ zi} hoisam.
 only.have Mary go party John SIN/ SINZI/ ZI happy
 ‘Only if Mary goes to the party will John be happy. (He is being so picky.)’

To conclude, we follow Wang (2025b) and others in taking *zi*, not *sin*, to have been the original carrier of the scalar flavor. Temporal *sin* acquired that flavor by absorbing the semantics of *zi*.⁹

While *sin_S* can be replaced by *zi* or *sin_Szi*, *sin_T* and *sin_{QF}* cannot.

- (20) Mary heoi-zo jatbun {sin_T/ *zi/ *sin_Tzi}.
 Mary go-PRV Japan SIN ZI SINZI
 ‘Mary went to Japan first.’
- (21) Mary jam m jam hoengban {sin_{QF}/ *zi/ *sin_{QF}zi}?
 Mary drink NEG drink champagne SIN ZI SINZI
 ‘Wait, does Mary drink champagne?’

If *sin_{QF}* had been further derived from *sin_S*, we would expect *sin_{QF}* to be at least partly interchangeable with *zi* or *sinzi*. It is not. Moreover, question-focus *sin* preserves the lexical meaning of *sin_T*. Temporal *sin* sequences two events; question-focus *sin* sequences two questions, as in (22). No such sequencing is present in the scalar cases, whether leftward (23) or rightward (24).

⁹This section is only a brief summary of the historical literature. Historical development is not the main concern of the thesis, so we refer the reader to Cai and Huang (2025), Li (2019), Wang (2025a) for full diachronic corpus studies. The recent consensus is that scalar *sin* derives from *zi* via an intermediate form *sinzi*. That is enough for present purposes: temporal *sin* derives scalar *sin* and question-focus *sin* separately, and the formal account is built accordingly.

- (22) A: Mary will drink champagne tonight.
 B: dangzan, Mary jam m jam zau sin?
 B: wait Mary drink NEG drink alcohol SIN
 B: ‘Wait, does Mary drink alcohol, first of all?’
 $\rightsquigarrow$ We should resolve *whether Mary drinks alcohol* first. Only then can we resolve *whether Mary will drink champagne tonight*.
- (23) John sin sik dak saam-go pinggwo.
 John SIN eat ONLY three-CL apple
 ‘It is little that John only ate three apples.’
 $\nrightarrow$ John only ate three apples first.
- (24) zijau John lai, Mary sin lai.
 only.have John come Mary SIN come
 ‘Only if John comes will Mary come.’
 $\nrightarrow$ Mary will come first, and then John will come.

We therefore propose a two-way grammaticalization. On one path, sin_T derives sin_S by absorbing zi , and the scalar flavor comes from zi . On the other path, sin_T derives sin_{QF} from its own lexical meaning.

1.3 The roadmap

Given the historical derivation in (25), Chapter 2 addresses (25-a), arguing that the QF use results from type-shifting sin_T from propositions to questions. Chapter 3 addresses (25-b), arguing that sin_S is a scalar operator that jointly works with an exclusive particle to presuppose a LONG distance between its prejacent p and a contextually salient expectation r . We further exploit the ambiguity of adverbial-focus sin_S to address the directionality puzzle for sin_S and its Mandarin counterpart *cai*. Chapter 4 turns to theoretical implications: extensions to Cantonese *zau*, comparison with Mandarin *cai* and *jiu*, the typology of scalar particles in Southern Sinitic, and cross-linguistic parallels with Vietnamese *mới*

and with various *only/until* particles in Romance and Pama-Nyungan languages. Chapter 5 collects remaining issues.

- (25) a. $\text{sin}_T \rightarrow \text{sin}_{QF}$
 b. $\text{sin}_T \rightarrow \text{sin}_S$

Chapter 2

From Temporal to QF

2.1 The temporal adverb *sin*

Temporal *sin_T* ‘first’ requires a distinct proposition to hold after the prejacents. Consequently (1) is felicitous, while (2) is not.

- (1) Mary lai party *sin_T*. John zihau lai
Mary come party SIN John then come
‘Mary will come to the party first. Then John will come.’
- (2) Mary lai party *sin_T*. #Jinhau mou laa4
Mary come party SIN then NEG SFP
‘Mary will come to the party first. Then nothing.’

The existence of that later event is a presupposition: it survives under negation, as in (3).

- (3) Mary m hai lai party sin. {keoi heoi sik maanfaan sin/ #Jinhau
Mary NEG COP come party SIN 3.SG go eat dinner SIN then
mou laa4.}
NEG SFP
Can mean: ‘Mary will not come to the party first. She will go to dinner

first.’

Cannot mean: ‘Mary will not come to the party first. And then nothing.’

The two propositions must also be distinct. (4) is unacceptable.

- (4) Mary lai party sin. #jinhau Mary lai (party).
 Mary come party SIN then Mary come party
 Int.: ‘Mary will come to the party first. Then Mary will come to the party.’

Thus sin_T sets up a scale of time points and presupposes that a distinct proposition is true at a later time. In (1), let p be *Mary coming to the party* and q be *John coming to the party*. Since p precedes q , p sits to the left of q on the time scale, as in Figure 2.1.

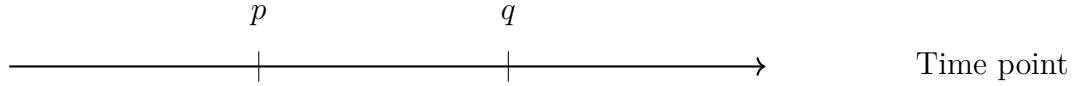


Figure 2.1: Temporal sin

Because the later proposition cannot be denied, we treat its existence as a presupposition and write (5).

$$(5) \quad \llbracket \text{sin}_{T\langle st, st \rangle} \rrbracket(C_i) = \lambda p_{\langle st \rangle} \lambda w : \exists q \in C_i [p <_{TP} q]. p(w)$$

, where $<_{TP}$ is a scale of time points.

‘Temporal sin is truth-conditionally vacuous, but presupposes that a distinct proposition is true at a later time.’

Applying (5) to (1) yields (6), where $p = \lambda w'. \text{COME}(m, \text{party}, w')$.

$$(6) \quad \llbracket \text{Mary lai party sin}_T \rrbracket(C_i) = \lambda w : \exists q \in C_i [p <_{TP} q]. \text{COME}(m, \text{party}, w)$$

2.2 QF *sin*: the distribution

Sin_{QF} is defined only if there is a current question Q_2 under discussion that the speaker cannot yet resolve. Only after the interlocutors resolve Q_1 —the prejacent of sin_{QF} —do they have enough information to resolve Q_2 . As Tang (2006) observes, sin_{QF} combines only with interrogatives. It is compatible with *wh*-interrogatives (7), alternative interrogatives (8) and A-not-A interrogatives (9), but not with echo interrogatives (10).

- (7) bingo heoi sin?
 who go SIN
 ‘Wait, who will go, first of all?’ [Tang 2006 ex. 7]
- (8) jam naaicaa ding gafe sin?
 drink milk.tea or coffee SIN
 ‘Wait, do you drink milk tea or coffee, first of all?’ [Tang 2006 ex. 8]
- (9) nei heoi m heoi sin?
 2.SG go NEG go SIN
 ‘Wait, will you go, first of all?’ [Tang 2006 ex. 9]
- (10) *bingo zeoi lang waa sin?
 who most beautiful SFP SIN
 Int.: ‘Wait, the most beautiful is who?’ [Tang 2006 ex. 12]

Symbesma (2013) argues that sin_{QF} is derived from postverbal temporal *sin*. It does not order two propositions, but two questions Q_1 and Q_2 . The more advanced question Q_2 is presupposed: sin_{QF} is infelicitous out of the blue, like question-focus *even* in English (Iatridou and Tatevosov, 2016). What kinds of content can serve as Q_2 is a separate question, which we address by comparing sin_{QF} with the sentence-final particle *me1*.

First, sin_{QF} is a poor way of targeting the addressee’s at-issue content, as in

(11).¹

- (11) A: Mary will drink champagne tonight.
 B: ??dangzan, Mary jam m jam hoengban sin?
 B: wait Mary drink NEG drink champagne SIN
 B int.: ‘Wait, does Mary drink champagne at all?’ [At-issue content]

To cast doubt on at-issue content, Cantonese uses the SFP *me1* instead, as in (12).

- (12) A: Mary will drink champagne tonight.

¹During the oral defense, Lawrence Cheung and Sze-Wing Tang suggested that *sin_{QF}* can target an assertion. Cheung offered (i), which he finds grammatical.

- (i) A: Aa-ming tingjat wui lei gaa.
 A: Ming tomorrow will come SFP
 A: ‘Ming will come tomorrow.’
 B: Aa-ming tingjat hai-mai wui lei *sin_{QF}*? Keoi ngaamngaam tung neoi
 B: Ming tomorrow be-NEG will come SIN 3.SG just.now with girl
 pengjau fen-zo sau wo.
 friend break-PRV hand SFP
 B: ‘Will Ming come? He just broke up with his girlfriend.’

Speaker variation is real: two consultants and I find (i) degraded. Acceptability improves if *zanhai* ‘really’ is added, as in (ii).

- (ii) A: Aa-ming tingjat wui lei gaa.
 A: Ming tomorrow will come SFP
 A: ‘Ming will come tomorrow.’
 B: Aa-ming tingjat hai-mai zanhai wui lei *sin_{QF}*? Keoi ngaamngaam tung neoi
 B: Ming tomorrow be-NEG really will come SIN 3.SG just.now with girl
 pengjau fen-zo sau wo.
 friend break-PRV hand SFP
 B: ‘Will Ming really come? He just broke up with his girlfriend.’

We leave the variation for further work.

B: dangzan, Mary jam hoengban me1?
 B: wait Mary drink champagne ME
 B: ‘Wait, does Mary drink champagne?’ [At-issue content]

Me1 conveys a negative bias and is incompatible with interrogative syntax, as in (13).

- (13) *bingo lai party me1?
 who come party SFP
 Int.: ‘Who came to the party?’

Second, *sin_{QF}* can target an entailment of the addressee’s utterance, as in (14). *Me1* can do the same, as in (15).

- (14) A: Mary will drink champagne tonight.
 B: dangzan, Mary jam m jam zau sin_{QF}?
 B: wait Mary drink NEG drink alcohol SIN
 B int.: ‘Wait, does Mary drink alcohol, first of all?’ [Entailment]
- (15) A: Mary will drink champagne tonight.
 B: dangzan, Mary jam zau me1?
 B: wait Mary drink alcohol ME
 B: ‘Wait, does Mary drink alcohol?’ [Entailment]

The difference is that *me1* encodes a negative bias toward the targeted content, whereas *sin_{QF}* is inquisitive: it marks the issue as one that should be resolved first, without linguistically encoding a bias.

Third, both particles can target a lexical presupposition, as in (16) and (17). Again, *me1* is biased and *sin_{QF}* is not.

- (16) A: Mary stopped smoking.
 B: dangzan, Mary jau mou sik-gwo jin sin_{QF}?
 B: wait Mary have NEG eat-EXP cigarette SIN

B int.: ‘Wait, has Mary ever smoked, first of all?’[Lexical presupposition]

- (17) A: Mary stopped smoking.
B: dangzan, Mary zicin sik jin me1?
B: wait Mary before eat cigarette ME
B int.: ‘Wait, did Mary use to smoke?’ [Lexical presupposition]

Fourth, both can target an implicature, as in (18) and (19).

- (18) A: Mary ate 3 apples. [Implies: Mary only ate 3 apples.]
B: dangzan, Mary hai m hai zinghai sik-zo saam-go pinggwo sin_{QF}?
B: wait Mary COP NEG COP only eat-PRV three-CL apple SIN
B: ‘Wait, did Mary only eat three apples?’ [Implicature]
- (19) A: Mary ate 3 apples. [Implies: Mary only ate 3 apples.]
B: dangzan, Mary zinghai sik-zo saam-go pinggwo me1?
B: wait Mary only eat-PRV three-CL apple ME
B: ‘Wait, did Mary only eat three apples?’ [Implicature]

Fifth, only *sin_{QF}* is licensed by a uniqueness presupposition, as in (20) (context from Theiler 2021). *Me1* is not, as in (21). Because B does not know which Anna is at issue, the prejacents would have to be interrogative; but *me1* takes only a declarative, so (21) is banned syntactically as well as semantically.

- (20) Context: there are two Annas who are friends with both A and B.
A: Anna called. [Presupposes: there is a unique Anna.]
B: dangzan, nei gong bingo Anna sin_{QF}?
B: wait, 2.SG say which Anna SIN
B: ‘Wait, which Anna did you refer to?’ [Uniqueness presupposition]
- (21) Context: there are two Annas who are friends with both A and B.
A: Anna called. [Presupposes: there is a unique Anna.]
B: *dangzan, nei gong bingo Anna me1?
B: wait, 2.SG say which Anna ME
B int.: ‘Wait, which Anna did you refer to?’ [Uniqueness presupposition]

Finally, sin_{QF} can target content that bears no entailment, presupposition or implicature relation to its prejacent. In (22), *whether Mary will come to the party* neither asserts, presupposes nor implicates *whether John will come*, but the former is a premise of the latter. Sin_{QF} is felicitous; me1 is not.

(22) Context: John likes Mary.

Ann: Hi John, will you come to the party tonight?

John: Mary jammaan lai m lai party sin_{QF} ?

John: Mary tonight come NEG come party SIN

‘Wait, will Mary come to the party, first of all?’

[Premise]

(23) Context: John likes Mary.

Ann: Hi John, will you come to the party tonight?

John: #Mary jammaan lai party me1 ?

John: Mary tonight come party SFP

Int.: ‘Wait, will Mary come to the party, first of all?’

[Premise]

The contrast is summarized in Table 2.1.

Table 2.1: Sin_{QF} and me1 as targeting devices

Targeted content	sin_{QF}	me1
At-issue assertion	?	✓
Entailment	✓	✓
Lexical presupposition	✓	✓
Implicature	✓	✓
Uniqueness presupposition	✓	*
Premise	✓	*

In short, sin_{QF} targets presuppositions, implicatures, entailments, uniqueness presuppositions and premises, but not assertions. Me1 targets lexical presuppositions, implicatures, entailments and assertions, but neither uniqueness presuppositions nor premises.

2.3 QF *sin*: the formulation

To state the semantics of sin_{QF} we first need a working semantics for me1 . Me1 is felicitous only if the speaker is negatively biased toward a proposition to which the addressee is committed.² That commitment may be an assertion, an entailment or a lexical presupposition. For present purposes we treat me1 as a negatively biased question particle, with the bias marked in **bold**, as in (24) (but cf. Hara 2014, Su 2025, Wakefield 2020 for finer-grained accounts).

$$(24) \quad \llbracket \text{me1}_{\langle st, stt \rangle} \rrbracket = \lambda p. \{ \lambda w. p(w), \lambda w. \neg \mathbf{p}(\mathbf{w}) \}$$

, where p is a proposition to which the addressee is committed.

‘ Me1 takes a proposition and returns a negatively biased polar question.’

This is why me1 can target assertions, lexical presuppositions, entailments and implicatures: all are propositional objects that an interlocutor can commit to and that a me1 -speaker can treat as false. It also explains the incompatibility with premises. In (22), Mary’s coming is a premise of John’s coming; but by asking whether John will come, Ann does not commit herself to anything about Mary. The premise is overlooked, not asserted as true. Me1 is therefore unavailable: from John’s point of view Ann has not said anything false. The uniqueness case

²On the ontology of commitments, see Farkas and Bruce (2010), Gunlogson (2001) and subsequent work. Here *commitment* is an umbrella term for any propositional content that is asserted or inferable from an utterance.

is similar. Speaker A says nothing about *which* unique Anna is at issue; she has simply failed to specify one. A has not said anything false; she has left something out. Sin_{QF} is a QUD navigator that puts the missing question on the table.

We can use the same ordering relation $<$ for the temporal use and the QF use. The difference is only in what is ordered: propositions versus questions ranked by necessary preconditionhood (cf. Theiler 2021).

(25) $Q_1 <_Q Q_2$ iff resolving Q_1 is a necessary precondition of resolving Q_2 .

The configuration is as in Figure 2.2.

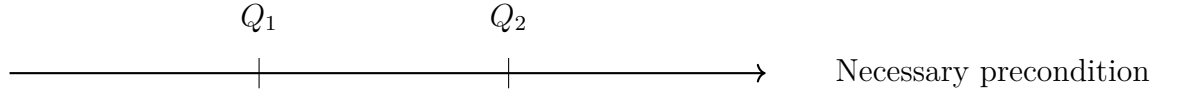


Figure 2.2: QF *sin*

Sin therefore type-shifts from propositions of type $\langle s, t \rangle$ to questions of type $\langle \langle s, t \rangle, t \rangle$, as in (26).

(26) $\llbracket \text{sin}_{QF} \rrbracket(C_i) = \lambda Q_{1\langle st, t \rangle} \lambda p_{\langle st \rangle} : \exists Q_2 [Q_2 \in C_i \wedge (Q_1 <_Q Q_2)]. Q_1(p)$
, where $<_Q$ is a scale of necessary preconditions.

‘QF *sin* is truth-conditionally vacuous, but presupposes a current Q_2 whose resolution depends on Q_1 .’

What is a necessary precondition? A necessary *condition* is the familiar relation in (27).

(27) P is a necessary condition of Q iff $Q \rightarrow P$.

If a bachelor is an unmarried adult male, being unmarried is a necessary condition of being a bachelor, not vice versa. If someone is not unmarried, the claim that

he is a bachelor is false. Necessary *preconditions* behave differently. In Strawson’s example (Strawson, 1950), *there is a king of France* is a necessary precondition of *the present king of France is bald*. The interlocutors must settle the former before the latter can even be discussed. If there is no king of France, *the present king of France is bald* is not false: the referential act never gets off the ground, so the question of truth or falsity does not arise. We therefore define the precondition scale as in (28).

- (28) $Q_{1\langle st,t \rangle}$ is a necessary precondition of $Q_{2\langle st,t \rangle}$ iff Q_1 must be settled before Q_2 can arise, be posed, or be evaluated.

This scale accounts for the distribution of $\sin_{QF}$. First, $\sin_{QF}$ cannot target assertive content, because a question cannot be its own necessary precondition: $Q_1 \neq Q_2$. Second, it can target lexical presuppositions, implicatures and entailments. In those cases the assertion is Q_2 and the inference is Q_1 ; Q_2 is discussable only if Q_1 is settled. If the inference fails, Q_2 is the wrong question to ask. If *whether Mary has ever smoked* is unsettled, *whether Mary stopped smoking* cannot be discussed. If *whether Mary drinks alcohol* is unsettled, *whether Mary drinks champagne* cannot be discussed. If *whether Mary only ate three apples* is unsettled, *whether Mary ate three apples* need not be the right question, because the obligatory scalar implicature of the latter may not hold (Chierchia, 2013). Third, unlike *me1*, $\sin_{QF}$ can target uniqueness presuppositions and premises, because those are precisely the regions of logical space that the addressee has overlooked. If *which unique Anna A is referring to* is unsettled, *whether (that unique) Anna called* is not well-formed. If the premise QUD *whether Mary will come* has not been addressed, the conclusion QUD *whether John will come* is not discussable.

Thus *me1* encodes a negative bias toward a commitment, while $\sin_{QF}$ promotes a more basic question relative to a more advanced one. They overlap on

lexical presuppositions, entailments and implicatures because those contents can either be treated as false or be treated as preconditions that make an assertion well-formed.

The type-shift from (5) to (26) is the formal counterpart of the historical claim in Chapter 1: sin_{QF} inherits the sequencing semantics of sin_T , replacing a scale of time points with a scale of necessary preconditions. No scalar LONG component is involved. That component belongs to the other grammaticalization path, to which we now turn.

Chapter 3

From Temporal to Scalar

3.1 Scalar *sin*: a PM account

This chapter gives a compositional semantics for scalar *sin* and the exclusive particles it co-occurs with: *dak*, *zikdou* and *zijau*. Previous accounts take the exclusive particle to have propositional scope (Bayer, 1996, 2012, 2019, Sun, 2021) and let the scalar particle scope over it by Functional Application (Hole, 2017, 2023, Yip, 2026), as in the LF in (1).

$$(1) \quad [\text{Scalar Op}_{\langle st, st \rangle} [\text{Exclusive Op}_{\langle st, st \rangle} [p_{\langle st \rangle}]]]$$

That LF does not work for *sin* on the analysis developed here. Both the scalar particle and the exclusive particle need to take the sentential proposition p as an argument. Because exclusive particles are not truth-conditionally vacuous, the scalar particle cannot take the original p ; it would have to take a proposition already modified by the exclusive. The alternative pursued here is that the two operators combine by Predicate Modification (Heim and Kratzer, 1998), as in (2).

$$(2) \quad \text{Predicate Modification (PM; after Heim \& Kratzer 1998)}$$

If α and β are of type $\langle \sigma, t \rangle$, $\llbracket \alpha \rrbracket = \langle \pi_\alpha, \varphi_\alpha \rangle$ and $\llbracket \beta \rrbracket = \langle \pi_\beta, \varphi_\beta \rangle$, where π is a presupposition and φ an assertion, then $\text{PM}(\llbracket \alpha \rrbracket, \llbracket \beta \rrbracket) = \langle \pi_\alpha \wedge \pi_\beta, \varphi_\alpha \wedge \varphi_\beta \rangle$.

Standard applications of PM conjoin asserted content. We assume that PM may also conjoin presuppositions, as the definition already allows. The resulting LF is (3).

$$(3) \quad [[\text{Scalar Op Exclusive Op}] [p]]$$

3.2 The *so-few* evaluation

Recall the *so-few* use of scalar *sin*, repeated as (4).

- (4) Context: John was expected to eat 10 apples, but...
 John sin_{few} sik-dak [saam] $_F$ -go pinggwo.
 John SIN eat-ONLY three-CL apple
 ‘John only ate three apples. (So few!)’

Sin yields the *so-few* evaluation by saying that the prejacent p^1 (*John sik saam-go pinggwo* ‘John ate three apples’) is far lower than a contextually salient expectation r (*John sik sap-go pinggwo* ‘John ate ten apples’). The configuration is Figure 3.1.

We take the exclusive particle *dak* to have the same meaning as English *only*, as in (5).

¹Strictly speaking, p is the prejacent of the *sin dak* complex assembled by PM, not of *sin* alone. We keep the shorter description for readability.

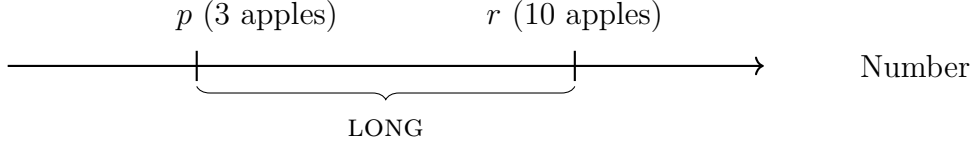


Figure 3.1: The *so-few* use

- (5) $\llbracket \text{dak}_{\langle st, st \rangle} \rrbracket(C_i) = \lambda p \lambda w : p(w). \neg \exists p' \in C_i [p <_N p' \wedge p'(w)]$
, where $<_N$ is a fully ordered number scale (Beaver and Clark, 2008).
‘*Dak* presupposes p and asserts that no stronger p' is true.’

(5) predicts that an addressee cannot deny (4) by asserting a weaker alternative (that alternative is presupposed), but can deny it by asserting a stronger alternative (that alternative is asserted to be false). Both predictions are borne out, as in (6) and (7).

- (6) A: John sin_{Few} sik-dak [saam]_F-go pinggwo.
B: John m hai sik-dak saam-go pinggwo. #Keoi sik-zo jat-go.
B: John NEG COP eat-ONLY three-CL apple. 3.SG eat-PRV one-CL
B int.: ‘John did not only eat three. He ate one.’
- (7) A: John sin_{Few} sik-dak [saam]_F-go pinggwo.
B: John m hai sik-dak saam-go pinggwo. Keoi sik-zo ng-go.
B: John NEG COP eat-ONLY three-CL apple. 3.SG eat-PRV five-CL
B: ‘John did not only eat three. He ate five.’

Dak is therefore an *only*. Matching Figure 3.1, we let sin_{Few} assert what *dak* asserts and add a *LONG* presupposition relating p to r , as in (8).

- (8) $\llbracket \text{sin}_{Few\langle st, st \rangle} \rrbracket(C_i) = \lambda p \lambda w : \exists r [p <_N r \wedge \text{LONG}(p, r)]. \neg \exists p' \in C_i [p <_N p' \wedge p'(w)]$
‘*Sin_{Few}* presupposes that p is far from an expectation r , and asserts that no stronger p' is true.’

As a presupposition, the LONG evaluation should survive under negation, in conditionals and in questions. That is confirmed by (9), (10) and (11).

- (9) Keoi m hai sin dak zunghok batjip! keoi jau PhD!
 3.SG NEG COP SIN only high.school graduate 3.SG have PhD
 ‘He is not only a high-school graduate! He has a PhD!’
 $\rightsquigarrow$ The speaker treats graduating from high school as far below the expectation.
- (10) Boss: jyugwo keoi sin_{Few} laai dak jat-go haak, zau caau-zo
 Boss: if 3.SG SIN pull ONLY one-CL customer, ZAU fire-PRV
 keoi.
 3.SG
 ‘If he has only persuaded one customer to use our service, fire him.’
 $\rightsquigarrow$ The boss treats persuading one customer as far below the expectation.
- (11) bingo sin sik-dak saam-go pinggwo?
 who SIN eat-ONLY three-CL apple
 ‘Who only ate three apples?’
 $\rightsquigarrow$ The speaker treats eating three apples as far below the expectation.

Relative to the general LF in (3), the *so-few* LF is (12).

- (12) LF for the *so-few* use: $[[\text{sin}_{Few} \text{dak}] [p]]$

(5) and (8) combine by Predicate Modification as in (13).

- (13) $\llbracket \text{sin}_{Few} \text{dak}_{\langle st, st \rangle} \rrbracket(C_i)$
 $= \lambda p \lambda w : p(w) \wedge \exists r [p <_N r \wedge LONG(p, r)]. \neg \exists p' \in C_i [p <_N p' \wedge p'(w)]$
 (the duplicated exclusive assertion is cancelled)

Sin_{Few} must assert what *dak* asserts so that PM conjoins two identical assertions and one copy can be dropped. If the two asserted different content, the leftover

assertion would strengthen the sentence beyond what is observed.²

Applied to the three-apple example, p is *John ate three apples*. The $\sin_{Few}$ *dak* complex presupposes that John ate three apples and that this is far from an expectation, and it asserts that John ate no more than three apples.

3.2.1 Speaker variation

Some speakers require *dak* ‘only’ for $\sin_{Few}$ to be grammatical; others allow $\sin_{Few}$ alone to carry both the scalar and the exclusive meaning, as in (14).³

- (14) %John $\sin_{Few}$ sik-zo [saam]_F-go pinggwo.
 John SIN eat-PRV three-CL apple
 ‘John only ate three apples. (So few.)’

The pattern is parallel to Mandarin cái_{Few} , which already encodes both the scalar and the exclusive meaning; an extra exclusive often feels redundant, as in (15).

²An alternative is to leave *dak* unchanged and let *sin* assert nothing, as in (i).

- (i) $\llbracket \sin_{Few\langle st, st \rangle} \rrbracket(C_i) = \lambda p \lambda w : \exists r[p <_N r \wedge LONG(p, r)].\emptyset$
 ‘*Sin* presupposes that p is far from an expectation r , and asserts nothing.’

(i) is cleaner, but an entry that asserts $\emptyset$ is unusual and raises a vacuous-binding problem. Both versions are compatible with the PM analysis.

³During the oral defense, Cheung noted that although *dak* often co-occurs with scalar *sin*, *dak* can be replaced by *zo* when the polarity-focus marker *hai* is present.

- (i) John $\sin_{Few}$ hai sik-zo saam-go pinggwo.
 John SIN COP eat-PRV three-CL apple
 ‘John ate three apples. (So few.)’

On that grammar, $\sin_{Few}$ itself is both exclusive and scalar, and the entry in (16) extends to it.

- (15) Yuehan cai_{Few} (?zhi) chi-le san-ge pingguo.
 John CAI only eat-PRV three-CL apple
 ‘John only ate three apples.’

For those speakers we assign *sin'*_{Few}—and Mandarin *cai*_{Few}—the conjoined entry in (16).

$$(16) \quad \llbracket \text{sin}'_{Few\langle st, st \rangle} \rrbracket(C_i) = \llbracket \text{cai} \rrbracket(C_i) \\
= \lambda p \lambda w : p(w) \wedge \exists r[p <_N r \wedge LONG(p, r)]. \neg \exists p' \in C_i[p <_N p' \wedge p'(w)]$$

3.2.2 Could Functional Application still work?

If the relative scope of the exclusive and the scalar is reversed, Functional Application becomes available, as in (17).

$$(17) \quad [\text{Exclusive Op} [\text{Scalar Op} [p]]]$$

A scalar operator can be truth-conditionally vacuous and contribute only a pre-supposition. After the scalar applies, *p* is still *p*, and the exclusive can apply to it. For the *so-few* use, *sin''*_{Few} is vacuous and *dak* is unchanged, as in (18)–(19).

$$(18) \quad \llbracket \text{sin}''_{Few\langle st, st \rangle} \rrbracket(C_i) = \lambda p \lambda w : \exists r[p <_N r \wedge LONG(p, r)]. p(w)$$

$$(19) \quad \llbracket \text{dak}_{\langle st, st \rangle} \rrbracket(C_i) = \lambda p \lambda w : p(w). \neg \exists p' \in C_i[p <_N p' \wedge p'(w)]$$

Functional Application then yields (20).

$$(20) \quad \llbracket \text{dak} \rrbracket(\llbracket \text{sin}''_{Few} \rrbracket(C_i)) = \lambda p \lambda w : p(w) \wedge \exists r[p <_N r \wedge LONG(p, r)]. \neg \exists p' \in C_i[p <_N p' \wedge p'(w)]$$

While this is technically possible, this LF is not supported by any literature on scalarity, to the best of my knowledge. We leave this approach for reference here.

3.3 The *so-late* evaluation

Recall the *so-late* use, as in (21).

- (21) Context: Mary should sleep at 12am, but...
 (zikdou) [tingwong]_F Mary *sin_{Late}* fangaau.
 not.until dawn Mary SIN sleep
 ‘Mary did not sleep until dawn. (So late!)’

Unlike *sin_{Few}*, which associates with a focus to its right, *sin_{Late}* associates with a focus to its left. We argue that reversing the direction of focus reverses the scale: $p < r$ for *sin_{Few}*, $r < p$ for *sin_{Late}*. The role of r changes with the scale. For *sin_{Few}*, r is an endpoint—the value the prejacent was expected to reach. For *sin_{Late}*, r is a starting point of observation, and the QUD alternatives are checkpoints along a path. In (21) the QUD is *When did Mary sleep?*, with alternatives a continuum of times. From the initial observation point r , a closer checkpoint p' (midnight) was expected to suffice for the consequent q (*Mary sleeps*); in fact a later p (dawn) is required. *Sin_{Late}* presupposes a LONG distance between p and r , as in Figure 3.2.

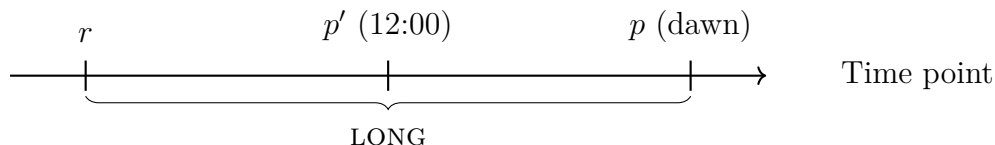


Figure 3.2: The sequential use

Three claims structure the LF. First, *sin_{Late}* co-occurs with *zikdou* ‘not until’. *Zikdou* may be phonologically covert, but it can always be spelled out; a *sin* that cannot host overt *zikdou* is not *sin_{Late}*. Second, while *dak* is a two-place operator on a proposition and a world, *zikdou* type-shifts to a three-place operator on two properties and an individual, or on two propositions and a world. Third,

sin_{Late} itself type-shifts between propositional and property arguments. (21) is a property case (one shared subject). A propositional case, with two subjects, is (22).

- (22) (zikdou) Mary fatlau John sin zi co.
 not.until Mary lose.temper John SIN know fault
 ‘It was not until Mary lost her temper that John admitted his fault. (So late!)’

The two LFs are (23) and (24).

- (23) sin_{Late} LF (property): $[[\text{sin}_{Late}\langle et, \langle et, et \rangle \rangle \text{zikdou}\langle et, \langle et, et \rangle \rangle] [P_{\langle et \rangle} Q_{\langle et \rangle}]]$

- (24) sin_{Late} LF (proposition): $[[\text{sin}_{Late}\langle st, \langle st, st \rangle \rangle \text{zikdou}\langle st, \langle st, st \rangle \rangle] [p_{\langle st \rangle} q_{\langle st \rangle}]]$

Following Hole (2004), we treat Cantonese *zikdou-sin* as parallel to English *not until*. English *not until* carries an actuality inference (AI) and a beyond-expectation inference (BEI) (Condoravdi, 2008, Declerck, 1995, Iatridou and Zeijlstra, 2021, Karttunen, 1974); BEI is the source of the *so-late* evaluation. The same inferences are found with *zikdou-sin_{Late}*, as in (25) and (26).

- (25) (zikdou) tingwong Mary sin fangaau. #Tingwong zihau Mary dou
 not.until dawn Mary SIN sleep dawn after Mary DOU
 mou fangaau.
 NEG sleep
 Int.: ‘Mary did not sleep until dawn. Mary did not sleep after dawn.’[AI]
- (26) (zikdou) tingwong Mary sin fangaau. #Tingwong fangaau hai houzhou.
 not.until dawn Mary SIN sleep dawn sleep COP early
 Int.: ‘Mary did not sleep until dawn. Sleeping at dawn counts as early.’
 [BEI]

The property and propositional entries for *zikdou* are (27) and (28).

$$(27) \quad \llbracket \text{zikdou}_{\langle et, \langle et, et \rangle} \rrbracket(C_i) = \lambda P_{\langle et \rangle} \lambda Q_{\langle et \rangle} \lambda x_e : P(x) \wedge Q(x). \neg \exists P'_{\langle et \rangle} \in C_i [P' <_{TP} P \wedge P'(x) \wedge Q(x)]$$

, where $<_{TP}$ is a scale of time points.

‘*Zikdou* presupposes that Q holds of x at the time P does, and asserts that Q was false of x at any earlier P' .’

$$(28) \quad \llbracket \text{zikdou}'_{\langle st, \langle st, st \rangle} \rrbracket(C_i) = \lambda p_{\langle st \rangle} \lambda q_{\langle st \rangle} \lambda w_s : p(w) \wedge q(w). \neg \exists p'_{\langle st \rangle} \in C_i [p' <_{TP} p \wedge p'(w) \wedge q(w)]$$

, where $<_{TP}$ is a scale of time points.

‘*Zikdou* presupposes that q is true when p is, and asserts that q was false at any earlier p' .’

The matching entries for sin_{Late} are (29) and (30).

$$(29) \quad \llbracket \text{sin}_{Late\langle et, \langle et, et \rangle} \rrbracket(C_i) = \lambda P \lambda Q \lambda x : \exists R_{\langle et \rangle} [R <_{TP} P \wedge LONG(R, P)]. \neg \exists P' \in C_i [P' <_{TP} P \wedge P'(x) \wedge Q(x)]$$

‘*Sin_{Late}* presupposes that P is far from a contextually salient earlier R , and asserts that Q was false at any time earlier than P .’

$$(30) \quad \llbracket \text{sin}'_{Late\langle st, \langle st, st \rangle} \rrbracket(C_i) = \lambda p \lambda q \lambda w : \exists r [r <_{TP} p \wedge LONG(r, p)]. \neg \exists p' \in C_i [p' <_{TP} p \wedge p'(w) \wedge q(w)]$$

‘*Sin_{Late}* presupposes that p is far from a contextually salient earlier r , and asserts that q was false at any time earlier than p .’

PM of (27) with (29), and of (28) with (30), yields (31) and (32).

$$(31) \quad \llbracket \text{sin}_{Late} \text{zikdou}_{\langle et, \langle et, et \rangle} \rrbracket(C_i) = \lambda P \lambda Q \lambda x : P(x) \wedge Q(x) \wedge \exists R [R <_{TP} P \wedge LONG(R, P)]. \neg \exists P' \in C_i [P' <_{TP} P \wedge P'(x) \wedge Q(x)]$$

$$(32) \quad \llbracket \text{sin}'_{Late} \text{zikdou}'_{\langle st, \langle st, st \rangle} \rrbracket(C_i) = \lambda p \lambda q \lambda w : p(w) \wedge q(w) \wedge \exists r [r <_{TP} p \wedge LONG(r, p)]. \neg \exists p' \in C_i [p' <_{TP} p \wedge p'(w) \wedge q(w)]$$

One advantage of this setup is that *zikdou* is parallel to *dak*: both presuppose their prejacent and assert a universal negation. They differ in which alternatives they negate. *Zikdou* negates all weaker (earlier) alternatives; *dak* negates all stronger ones. That difference is the source of the directionality puzzle, taken up in §3.5.

A second advantage is that the scalar contribution of *sin* is uniform: a LONG presupposition. The exclusive contribution is copied from the co-occurring exclusive particle.

The assertion of *sin_{Late}* can be denied by claiming that Mary slept earlier than dawn, as in (33).

- (33) Context: Mary should sleep at 12am, but..
 m.hai zikdou [tingwong]_F Mary sin_{Late} fangaau. Mary
 NEG.COP not.until dawn Mary SIN sleep Mary
 lingsan jatdim jiging faan-zo.
 early.morning one.o'clock already sleep-PRV
 ‘It is not the case that Mary did not sleep until dawn. Mary slept at 1am.’

AI and BEI, by contrast, are presuppositional. AI cannot be denied by another speaker unless the presupposition-targeting adverbial *ganbun* ‘at all’ is used, as in (34). BEI cannot be denied even with *ganbun*, as in (35).

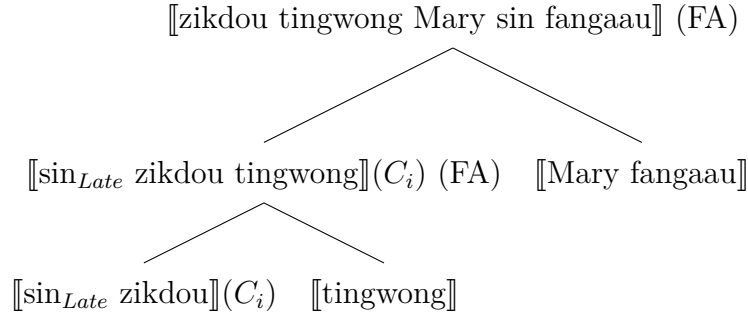
- (34) A: (zikdou) tingwong Mary sin fangaau.
 A: not.until dawn Mary SIN sleep
 A: ‘Mary did not sleep until dawn.’
 B: mhai. Mary #(ganbun) mou fangaau.
 B: NEG.COP Mary at.all NEG sleep
 B: ‘No. Mary did not sleep at all.’ [AI]
- (35) A: (zikdou) tingwong Mary sin fangaau.
 A: not.until dawn Mary SIN sleep
 A: ‘Mary did not sleep until dawn.’

B: m.hai. #(ganbun) tingwong sin faan hai houzhou ze1.
 B: NEG.COP at.all dawn SIN sleep COP very.early SFP
 B: ‘No. It is early to sleep at dawn.’

AI is the presupposition contributed by the exclusive: actualization of P forces actualization of Q . BEI is the presupposition contributed by the scalar: a LONG distance between R and P means that Q was expected at an earlier P' .

The property case involves a non-trivial type shift. Following Wimmer (2022a), we take the properties in (21) to be $\lambda x_e.\text{dawn}(x)$ and $\lambda x_e.\text{sleep}(m, x)$: *sleep* is shifted so as to take an extra individual argument. The composition is (36)–(38).

(36) LF for (21)



(37) Ingredients for (21)

- a. $\llbracket \text{sin}_{Late} \text{ zikdou} \rrbracket (C_i) = \lambda P \lambda Q \lambda x : P(x) \wedge Q(x) \wedge \exists R [R <_{TP} P \wedge LONG(R, P)]. \neg \exists P' \in C_i [P' <_{TP} P \wedge P'(x) \wedge Q(x)]$
- b. $\llbracket \text{tingwong} \rrbracket = \lambda y.\text{dawn}(y)$
- c. $\llbracket \text{Mary fangaau} \rrbracket = \lambda z.\text{sleep}(m, z)$

(38) Composition of (21)

- a. $\llbracket \text{zikdou tingwong sin}_{Late} \rrbracket (C_i) = \llbracket \text{sin}_{Late} \text{ zikdou} \rrbracket (\llbracket \text{tingwong} \rrbracket) (C_i) = \lambda Q \lambda x : \text{dawn}(x) \wedge Q(x) \wedge \exists R [R <_{TP} \lambda y.\text{dawn}(y) \wedge LONG(R, \lambda y.\text{dawn}(y))]. \neg \exists P' \in C_i [P' <_{TP} \lambda y.\text{dawn}(y) \wedge P'(x) \wedge Q(x)]$
- b. $\llbracket \text{zikdou tingwong Mary sin fangaau} \rrbracket (C_i) = \lambda x : \text{dawn}(x) \wedge \text{sleep}(m, x) \wedge$

$$\begin{aligned} & \exists R[R <_{TP} \lambda y.\text{dawn}(y) \wedge \text{LONG}(R, \lambda y.\text{dawn}(y))]. \\ & \neg \exists P' \in C_i[P' <_{TP} \lambda y.\text{dawn}(y) \wedge P'(x) \wedge \text{sleep}(m, x)] \end{aligned}$$

The propositional case involves no comparable type shift; we omit the derivation.

3.4 The *so-picky* evaluation

The *so-picky* example is repeated as (39).

- (39) A: Beth is coming. Will you come?
 B: (zijau) [Mary]_F lai, ngo sin lai.
 B: only.if Mary come, 1.SG SIN come
 B: ‘Only if Mary comes will I come.’

We treat *sin_{Picky}* as parallel to *sin_{Late}*, with small adjustments.⁴

(39) says that although a weaker condition p' (*Beth coming*) was expected to suffice for q (*B coming*), only a stronger condition p (*Mary coming*) does. The path from the initial point r to the antecedent p is therefore long, as in Figure 3.3.

Every instance of *sin_{Picky}* co-occurs with *zijau*. *Zijau* may be covert, but it must be spellable; a *sin* that cannot host overt *zijau* is not *sin_{Picky}*. The LF is (40).

⁴The parallel between *zijau-sin_{Picky}* and *zikdou-sin_{Late}* has cross-linguistic support. Vietnamese *mói* is closely similar to scalar *sin*. Its modal use is ambiguous between a *not-until* reading and an *only-if* reading, as in (i).

- (i) Co bien lai moi_{MOD} lay duoc so-mi.
 exist receipt only-then take get shirt
 ‘You can’t get your shirt until you produce the receipt.’
 i. ‘You must produce the receipt in order to get the shirt.’
 ii. ‘Only if the receipt exists will you get the shirt.’ [Tran and Zimmermann 2021 ex. 2]

Vietnamese is taken up again in Chapter 4.

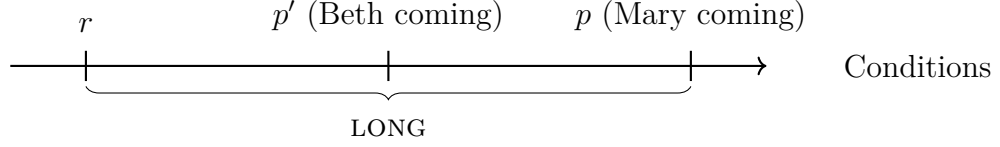


Figure 3.3: The picky use

$$(40) \quad \text{sin}_{Picky} \text{ LF: } [[\text{sin}_{Picky} \text{ zijau}] [p \ q]]$$

We adapt Wimmer’s (2022) entry for Mandarin *zhìyou* to Cantonese *zijau*, as in (41).

$$(41) \quad \llbracket \text{zijau}_{\langle st, \langle st, st \rangle} \rrbracket(C_i) = \lambda p \lambda q \lambda w : \exists w' \in \text{Acc}(w)[p(w') \wedge q(w')]. \neg \exists p' \in C_i[p' <_C p \wedge \exists w''[p'(w'') \wedge q(w'')]]$$

, where $<_C$ is a scale of conditions.

‘It is possible that p and q both hold; no weaker condition than p makes q possible.’

Zijau differs from *zikdou* in two ways. First, it quantifies over accessible worlds w' and w'' , as expected for conditionals; see Wimmer (2022a) for the full argument. Second—and more important here—the alternatives are ranked on a scale of conditions rather than time points. The higher a condition, the pickier it is. In the running example, B is introverted and goes to a party only if friends do. Beth is a good friend and Mary the best friend, so *Mary going* outranks *Beth going* on an intimacy scale. B is picky enough that only a friend as close as Mary will do.⁵

⁵An anonymous reviewer suggested ranking conditions by likelihood instead (Erlewine, 2014, Horn, 1969, Rooth, 1985). That would flip the scale: p (*Mary coming*) would sit below r (*Beth coming*), unifying the directionality of sin_{Few} and sin_{Picky} as ‘ p far below r ’. We reject the suggestion. *Mary coming* and *Beth coming* are independent events with independent likelihoods; (39) remains good even if Mary is the more likely guest. What is stable is that Mary is a closer

$\text{Sin}_{\text{Picky}}$ again contributes a LONG presupposition and copies its assertion from the exclusive, as in (42).

$$(42) \quad \llbracket \text{sin}_{\text{Picky}\langle st, \langle st, st \rangle \rangle} \rrbracket(C_i) = \lambda p \lambda q \lambda w : \exists r[r <_C p \wedge \text{LONG}(r, p)]. \neg \exists p' \in C_i[p' <_C p \wedge \exists w''[p'(w'') \wedge q(w'')]]$$

‘The condition p is much stronger than expected; no weaker condition makes q possible.’

PM of (41) and (42) yields (43).

$$(43) \quad \llbracket \text{sin}_{\text{Picky}} \text{ zijau}_{\langle st, \langle st, st \rangle \rangle} \rrbracket(C_i) = \lambda p \lambda q \lambda w : \exists w' \in \text{Acc}(w)[p(w') \wedge q(w')] \wedge \exists r[r <_C p \wedge \text{LONG}(r, p)]. \neg \exists p' \in C_i[p' <_C p \wedge \exists w''[p'(w'') \wedge q(w'')]]$$

As with sin_{Late} , $\text{sin}_{\text{Picky}}$ type-shifts to properties, in the subject-focus use (44) and the fronted-object use (45) (Sun, 2021).⁶

$$(44) \quad \begin{array}{l} \text{zijau} \quad [\text{John}]_F \text{ sin hoji cinggau sai} \quad \text{gaai!} \\ \text{only.have 2.SG} \quad \text{SIN can save} \quad \text{world} \\ \text{‘Only John can save the world (, which is a demanding task).’} \end{array}$$

friend than Beth, so the pickiness scale must be kept. Given the parallel with sin_{Late} , whose time-point scale cannot be inverted, $\text{sin}_{\text{Picky}}$ and sin_{Late} should both mean that p is far *above* r .

⁶For (44), the properties are *John* and *save the world*, respectively. That is, John must type-shift from type e to type $\langle e, t \rangle$. That is, a property of *being John*. Yenan Sun offers (i) to support the type-shifting of *John*.

$$(i) \quad \begin{array}{l} \text{zijau} \quad \text{John} \{ \text{nei-zung} \quad \text{jan}/^{??} \text{nei-go jan} \} \quad \text{sin}_{\text{Picky}} \text{ hoji cinggau} \\ \text{only.have John this-CL.KIND person} \quad \text{this-CL.INDIVIDUAL person SIN can} \\ \text{saaigai.} \\ \text{save} \quad \text{world} \\ \text{‘Only people like John can save the world.’} \\ \text{NOT: ‘Only this person, John, can save the world.’} \end{array}$$

- (45) [zijau [ngaujuk]_F]₁ Mary sin_{Picky} sik t₁.
 only.have beef Mary SIN eat t₁
 ‘Only beef does Mary eat. (She is so picky!)’

The fronted-object case is not the same *sin* as in-situ object focus (46).

- (46) Context: John was expected to eat 10 apples, but...
 John sin_{Few} sik-dak [saam]_{F-go} pinggwo.
 John SIN eat-ONLY three-CL apple
 ‘John only ate three apples. (So few!)’

Tang (2000) observes that in-situ *sin_{Few}* is subject to a cardinality effect: the object must be [+Num], and Num⁰ must bear prosodic focus. Hence (47) is degraded.

- (47) Context: John was expected to eat 10 apples, but...
 ??John sin_{Few} sik-dak pinggwo.
 John SIN eat-ONLY apple
 Int.: ‘John only ate apples. (So few!)’

Fronted-object focus, by contrast, is fine with [−Num] DPs, as in the beef example.⁷

- (48) *[zijau saam-faai ngaujuk]₁ Mary sin_{Picky} sik t₁.
 only.have three-CL beef Mary SIN eat t₁

⁷Yenan Sun offers (i).

- (i) [zijau sap-faai ngaujuk]_F sin gau John sik t₁
 only ten-CL beef SIN suffice John eat
 ‘Only ten pieces of beef can make John full. (John is so hungry!)’

The cardinality restriction is therefore confined to in-situ *sin_{Few}*.

Int.: ‘Only three pieces of beef does Mary eat. (She is so picky!)’

Following [Sun and Ng \(Forthcoming\)](#), we treat the contrast as a difference in the question answered. In-situ object focus answers a quantity question (*How many apples did John eat?*); the answer 3 is lower than expected. Fronted-object focus answers a quality question (*How picky is Mary’s diet?*); the answer is higher than expected. The two object-focus patterns therefore involve different scalar *sins*.

The property-level entries, again after [Wimmer \(2022a\)](#), are (49) and (50).

$$(49) \quad \llbracket \text{zijau}'_{\langle et, \langle et, et \rangle \rangle} \rrbracket(C_i) = \lambda P \lambda Q \lambda x : \exists x [P(x) \wedge Q(x)]. \neg \exists P' \in C_i [P' <_C P \wedge \exists y [P'(y) \wedge Q(y)]]$$

, where $<_C$ is a scale of conditions.

‘It is possible that P and Q both hold of some x ; no weaker condition than P makes Q possible.’

$$(50) \quad \llbracket \text{sin}'_{\text{Picky}_{\langle et, \langle et, et \rangle \rangle}} \rrbracket(C_i) = \lambda P \lambda Q \lambda x : \exists R [R <_C P \wedge \text{LONG}(R, P)]. \neg \exists P' \in C_i [P' <_C P \wedge \exists y [P'(y) \wedge Q(y)]]$$

PM yields (51).

$$(51) \quad \llbracket \text{sin}'_{\text{Picky}} \text{zijau}'_{\langle et, \langle et, et \rangle \rangle} \rrbracket(C_i) = \lambda P \lambda Q \lambda x : [P(x) \wedge Q(x)] \wedge \exists R [R <_C P \wedge \text{LONG}(R, P)]. \neg \exists P' \in C_i [P' <_C P \wedge \exists y [P'(y) \wedge Q(y)]]$$

The fronted-object composition is (52)–(53).

$$(52) \quad \text{LF of (45)}$$

$$\begin{array}{c}
\llbracket \text{zijau}' \text{ ngaujuk Mary sin}'_{Picky} \text{ sik} \rrbracket (C_i) \text{ (FA)} \\
\swarrow \quad \searrow \\
\llbracket \text{zijau}' \text{ ngaujuk sin}'_{Picky} \rrbracket (C_i) \text{ (FA)} \quad \llbracket \text{Mary sik} \rrbracket \\
\swarrow \quad \searrow \\
\llbracket \text{sin}'_{Picky} \text{ zijau}' \rrbracket (C_i) \quad \llbracket \text{ngaujuk} \rrbracket
\end{array}$$

(53) Composition of (45)

a. $\llbracket \text{zijau}' \text{ ngaujuk sin}'_{Picky} \rrbracket (C_i) = \llbracket \text{sin}'_{Picky} \text{ zijau}' \rrbracket (\llbracket \text{ngaujuk} \rrbracket) (C_i)$
 $= \lambda Q \lambda x : [\text{beef}(x) \wedge Q(x)] \wedge \exists R [R <_C \lambda a_e. \text{beef}(a) \wedge \text{LONG}(R, \lambda a_e. \text{beef}(a))].$
 $\neg \exists P' \in C_i [P' <_C \lambda a_e. \text{beef}(a) \wedge \exists y [P'(y) \wedge Q(y)]]$

b. $\llbracket \text{zijau}' \text{ ngaujuk Mary sin}'_{Picky} \text{ sik} \rrbracket (C_i) = \lambda x : [\text{beef}(x) \wedge \text{eat}(m, x)] \wedge$
 $\exists R [R <_C \lambda a_e. \text{beef}(a) \wedge \text{LONG}(R, \lambda a_e. \text{beef}(a))].$
 $\neg \exists P' \in C_i [P' <_C \lambda a_e. \text{beef}(a) \wedge \exists y [P'(y) \wedge \text{eat}(m, y)]]$

Subject focus works in the same way, with $P(x)$ and $Q(x)$ the subject and the predicate.⁸

⁸Cheung noted that (i) might seem to threaten the LONG intuition, because 89 marks is close to 90.

- (i) Context: one needs 90 marks out of 100 for a scholarship.

John sin haau 89 fan. ngo m hoji bei joenghokgam keoi.
 John SIN obtain 89 mark 1.SG NEG can give scholarship 3.SG
 ‘John only got 89 marks. I cannot give him a scholarship.’

The example does not threaten the analysis. 89 is the weaker alternative p' and 90 the stronger alternative p . Sin_{Picky} does not require p' and p to be far apart; it requires the distance from r to p to be long relative to the distance from r to p' . The antecedent need only outrank its contextually salient competitor.

3.5 The directionality puzzle

The pattern is now this. If the prosodic focus lies to the right of *sin* (*sin_{Few}*), the preajacent is lower than expected. If the prosodic focus lies to the left (*sin_{Late}*, *sin_{Picky}*), the relevant condition is higher than expected. What produces the asymmetry?

Adverbial-focus *sin* is the key. The adverb *aamaam* ‘just now’ may appear on either side of *sin* without an obvious change in meaning, as in (54).

- (54) Mary {[aamaam]_F sin/ sin [aamaam]_F} lai.
 Mary just.now SIN SIN just.now come
 Reading A: ‘Mary has only been here for a very short time. (So short!)’
 Reading B: ‘It was not until just now that Mary came. (So late!)’

Reading A measures time as duration and answers *How long has Mary been here?*. The sentence asserts that she has been here for only a short interval (say five minutes), presupposes an expected longer interval (say sixty minutes), and evaluates the gap as LONG. That is *sin_{Few}*. Reading B measures time as points and answers *When did Mary arrive?*. If now is 19:05 and ‘just now’ is 19:00, while arrival was expected at 18:00, the speaker marks the gap between 19:00 and 18:00 as LONG. A shorter duration of *p* being true corresponds to a later time point at which *p* becomes true. That is *sin_{Late}*. The two scales are mirror images of the same situation, as in Figures 3.4 and 3.5.

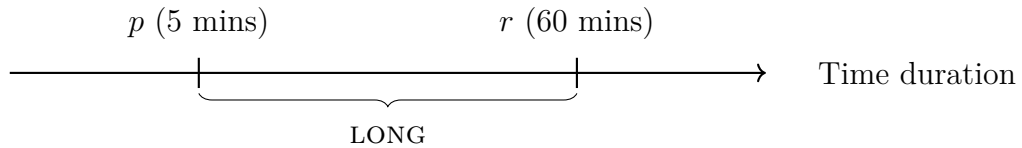


Figure 3.4: A time-duration reading for *sin*

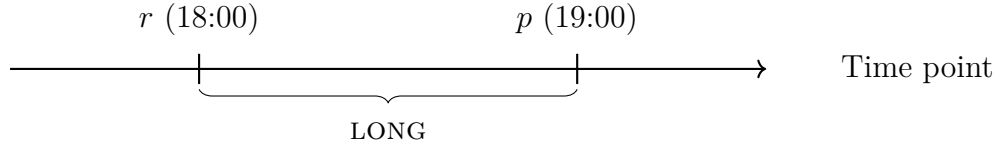


Figure 3.5: A time-point reading for *sin*

Because *aamaam* can sit on either side of *sin*, it can associate with either scale. The prediction is that left-adjoined *aamaam* favours the *so-late* reading, and right-adjoined *aamaam* the *so-few* (short-duration) reading. That is confirmed by the exclusive particles that each order licenses. Only left-adjoined *aamaam* co-occurs with *zikdou*, as in (55).⁹

- (55) Mary *zikdou* {*[aamaam]_F sin_{Late} / *sin_{Few} [aamaam]_F*} *lai*.
 Mary not.until just.now SIN SIN just.now come
 Unavailable: ‘Mary has only been here a short time. (So short!)’
 Only: ‘It was not until just now that Mary came. (So late!)’

Only right-adjoined *aamaam* co-occurs with *zinghai* ‘only’, as in (56).

- (56) Mary {**zinghai [aamaam]_F sin_{Late} / sin_{Few} zinghai [aamaam]_F*} *lai*.
 Mary only just.now SIN SIN just.now only come
 Unavailable: ‘It was not until just now that Mary came. (So late!)’
 Only: ‘Mary has only been here a short time. (So short!)’

Zinghai forces a *so-few* reading and therefore only co-occurs with *sin_{Few}*.

The directionality puzzle, then, is not a fact about *sin* having two lexical entries. It is a fact about which exclusive *sin* combines with, and about whether the associated scale measures ‘how little’ or ‘how late / how strong’. A single

⁹In the absence of *zikdou*, *aamaam sin* and *sin aamaam* need not differ; both orders can support both readings.

LONG presupposition covers both, once the exclusive—and the scale it induces—is factored out.

Chapter 4

The Sinitic family and beyond

4.1 Cantonese *zau*

Scalar *sin* cannot be understood in isolation from its counterpart *zau*, the analogue of the Mandarin *cai–jiu* contrast. The most conservative use of *zau* is the adverb ‘almost’, as in (1). Call this *zau_{Almost}*.

- (1) John *zau* sik jyun faan laa4.
John ZAU eat finish rice SFP
‘John almost finished his meal.’

Zau_{Almost} locates the prejacent below a contextually salient expectation. Unlike *sin*, however, it marks the gap as SHORT, as in Figure 4.1.

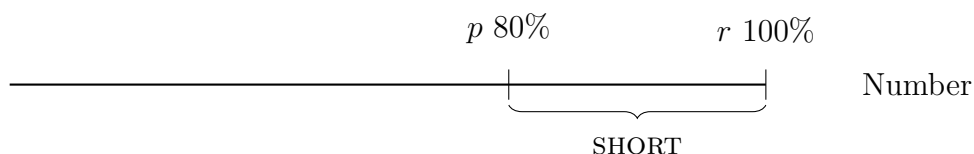


Figure 4.1: Adverbial *zau*

A second use conveys earliness and is parallel to *sin_{Late}*. Call this *zau_{Early}*.

Just as *sin_{Late}* co-occurs with *zikdou*, *zau_{Early}* co-occurs with elements such as *jat* ‘once’ (2) or *dou mei* ‘not even’ (3).¹

- (2) tin *([jat]_F) gwong Mary zau heisan.
 sky one light Mary ZAU get.up
 ‘Mary gets up once the sun rises.’
- (3) tin *(dou [mei]_F) gwong Mary zau heisan.
 sky DOU yet light Mary ZAU get.up
 ‘Mary gets up even before the sun rises.’

Zau therefore shares the directionality puzzle of scalar *sin*: $p < r$ for *sin_{Few}* and *zau_{Almost}*; $r < p$ for *sin_{Late}*, *zau_{Early}* and the conditionals below. When the scale flips, r changes from an endpoint to a starting point of observation. (2) is represented as Figure 4.2.

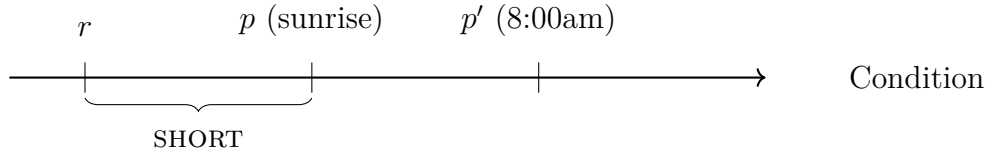


Figure 4.2: Early *zau*

Finally, *zau* co-occurs with *ziji* ‘as long as’ to mark minimal sufficiency, parallel to Mandarin *zhīyào-jīu* (Wimmer, 2022a,b). The context suggests that a stronger condition p' may be required for q ; the *ziji*-*zau* speaker asserts that a weaker p already suffices, as in (4). Call this *zau_{Easy}*. *Ziji* may be covert, but it can always be spelled out; a *zau* that cannot host overt *ziji* is not *zau_{Easy}*.

- (4) A: John’s best friend Mary will come to the party. Will John come?
 B: ziji John jau [jamho]_F pangjau lai party, J. *zau_{Easy}* wui
 B: only.need John have any friend come party, J. ZAU will

¹*Dou* is itself a much-discussed Sinitic particle; see Liu (2021b), Xiang (2008, 2020) on Mandarin *dou*.

lai.

come

B: ‘As long as any of John’s friends come to the party, John will come.’

Like *sin_{Picky}*, *zau_{Easy}* also takes properties, as in (5) and (6).

(5) A: We need a professor of engineering to solve the problem!

B: m sai. [John]_F zau gaau-dak dim laa1.

B: NEG need John ZAU solve-DAK work SFP

B: ‘No need. John can already solve the problem.’

(6) A: I only have Maltesers. Will Mary like them?

B: zijiu hai [tong]_F, Mary zau zungji sik.

B: only.need COP candy Mary ZAU like eat

B: ‘So long as it is candy, Mary will love to eat it.’

(4) is Figure 4.3: a longer path from r to p' was expected, but only the shorter path from r to p is required.

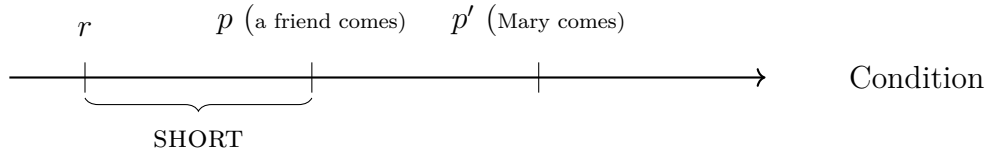


Figure 4.3: Conditional *zau*

Scalar *sin* and *zau* therefore share a directionality puzzle, and both depend on a co-occurring (often exclusive) element. An analysis of *sin* should constrain the analysis of *zau*.

4.2 Mandarin *cai* and *jiu*

Mandarin *cai* does not take an extra *only*, as in (7).

- (7) Mali cai (??zhi) chi-le san-ge pinggwo.
 Mali CAI only eat-PRV three-CL apple
 ‘Mary just ate three apples.’

Cai therefore does the exclusive work and the scalar work at once, as in (8).

- (8) $\llbracket \text{cai} \rrbracket(C_i) = \lambda p \lambda w : p(w) \wedge \exists r \in C_i [p < r \wedge \text{LONG}(p, r)]. \neg \exists p' [p < p' \wedge p'(w)]$

The *so-late* use is (9). *Cai* requires (possibly covert) *zhidao* ‘not until’, exactly as in Cantonese.

- (9) (zhidao) yanchanghui [jieshu]_F-le, Mali cai lai.
 not.until concert end-PRV Mary CAI come
 ‘It was not until the concert ended that Mary came.’

The *so-picky* uses (10)–(12) match *sin_{Picky}*.

- (10) zhiyou [Mali]_F lai, Yuehan cai lai.
 only.have Mary come John CAI come
 ‘Only if Mary comes will John come.’
- (11) zhiyou [heniu]_F Yuehan cai chi.
 only.have Wagyu John CAI eat
 ‘Only Wagyu does John eat.’
- (12) zhiyou [Mali]_F cai keyi jiejie zhe-ge wenti.
 only.have Mary CAI can solve this-CL problem
 ‘Only Mary can solve this problem.’

Adverbial focus, finally, associates in either direction (13).

- (13) Yuehan {cai [ganggang]_F / [ganggang]_F cai} lai.
 John CAI just.now just.now CAI come
 Duration reading: ‘John has only been here for a very short time.’
 Time-point reading: ‘It was not until just now that John came.’

The analysis of *sin_S* can therefore be adopted for *cai* with only local adjustments.

We do not extend the formal analysis to Mandarin *jiu*, but the parallels with *zau* are clear. As an adverb modifying AuxP, *jiu* means ‘almost’, like *zau_{Almost}* (14).

- (14) wo jiu yao zuo-hao le.
 1.SG JIU need do-PRV SFP
 ‘I have almost got it done.’

Jiu_{Early}, however, differs from *zau_{Early}*. In Mandarin the early use need not rely on *yi* ‘once’ or *haiwei* ‘not even’ (15). Why the two languages diverge here is an open question.

- (15) tian {yi/ haiwei/ Ø} liang Mali jiu qilai le.
 sky once even.yet Ø light Mary JIU get.up SFP
 Ø: ‘Mary gets up when the sun rises. (So early!)’
 yi: ‘Mary gets up once the sun rises. (So early!)’
 haiwei: ‘Mary gets up even before the sun rises. (So early!)’

Mandarin *zhiyao-jiu*, finally, is fully parallel to Cantonese *zijiū-zau* (16).

- (16) A: I only have Maltesers. Will Mary like them?
 B: zhiyao shi [tang]_F, Mali *(jiu) xihuan chi.
 B: only.need COP candy Mary JIU like eat
 B: ‘So long as it is candy, Mary will love to eat it.’

4.3 Southern Sinitic

Historically, Cantonese *sin* continues *zing³*. Li and Wu (2020) and others note that *zheng* (𪔐) is the most common exponent of this semantics across Southern Sinitic. Unlike northern *cai*, which is almost exclusively scalar, the Southern

exponents are multifunctional. Beyond the uses already discussed, three further uses occur.

First, the morpheme can be an adjective meaning ‘right’ or ‘good’, as in Lianjiang Cantonese (17)–(18).

- (17) 行得 正，企得 正。
walk-DAK ZING stand-DAK ZHENG
‘Walk properly, stand properly.’ [Lianjiang Cantonese]
- (18) 牛肉丸 好 正 啊。
beef.ball very ZHENG SFP
‘The beef balls are excellent!’ [Lianjiang Cantonese]

Second, it can be a contrastive adverbial, like Sinitic *pianpian* (偏偏) (Shi, 2023): the prejacent runs against the speaker’s expectation (19).

- (19) 喊 渠 走 ti24 邊，渠 正 走 ka24 邊 哦。
call 3.SG walk this.side 3.SG ZHENG walk that.side SFP
‘We told her to walk this way, but she walked that way.’ [Longgang Hakka]

Li and Wu (2020) also treat (20) as contrastive. It does not say that the speaker’s being able to see is unexpected; it acknowledges the prejacent and presupposes a *but q* that the addressee is invited to consider, like Sinitic *daoshi* (倒是).

- (20) 看 正 看 得到。
see ZING see able
‘I *can* see. (But...)’ [Pingjiang Gan]

Third, *zheng* is a degree adverb ‘very’ in some varieties (21).

- (21) 昨晡日 個 天氣 正 唔 好。
yesterday CL weather ZHENG NEG good
‘The weather yesterday was very bad.’ [Longjiang Hakka]

An emphatic use, highlighting polarity, is illustrated in (22)–(23). (The examples are Cantonese; the same pattern exists in Mandarin with *cai*.)

- (22) A: It’s cold outside!
 B: ceotmin sin [m]_F dung aa3!
 B: outside SIN NEG cold SFP
 B: ‘It’s NOT cold outside!’ [Cantonese]
- (23) A: New York is crowded!
 B: [hoenggong]_F sin bik aa3!
 B: Hong.Kong SIN crowded SFP
 B: ‘HONG KONG is crowded! (Not New York.)’ [Cantonese]

Li and Wu’s (2020) typology is Table 4.1.

Li and Wu (2020) observe that object focus (our *so-few* use), the conditional (our *so-picky* use) and the emphatic use are shared by every variety in the table. Mandarin *cai* has just those three plus the sequential (*so-late*) use, and none of the others. Those four uses are therefore a plausible common core. At least three other morphemes interact with that core: *xian* ‘first’, *zheng* ‘precisely’ and *zai* ‘then’.

No variety has the QF use without the temporal-adverb use, so the temporal use is a necessary precondition of QF. That supports the claim that QF grammaticalizes from the temporal adverb, with Cantonese *sin* as the case study. Cantonese *sin* itself lacks the contrastive and degree uses; those, we suggest, come from *zheng* and are left for further work. *Zai* ‘then’ is the source of the sequential use, which only some varieties have—plausibly via the well-known path from event sequencing to condition marking (Heine, 2002). Once sequential, the marker can acquire a conditional use and, from there, the rest of the *cai* core.

On this view the Southern multifunctionality is not a single linear grammaticalization of a *cai* core. It is the result of contact between that core and other

Table 4.1: Exponents of Cantonese *sin* across Sinitic varieties (Li and Wu, 2020)

	Hakka	Gan	Gan	Gan	Xiang	Hakka	Hakka	Hakka	Hakka	Longchuan	Lianjiang	Yue	Yue	Xiang
	Liancheng	Jishui	Pingxiang	Miluo	Ninghua	Meixian	Longchuan	Lianjiang	Beiliu	Yueyang				
Adjective	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Adverbial focus	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Object focus	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Conditional	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Sequential	–	–	–	–	+	+	+	+	+	+	+	+	+	+
Emphatic	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Temporal adverb	–	–	+	+	–	–	+	+	+	+	+	+	+	+
QF		–	+	+	–	–		+	–	+	+	+	+	+
Contrastive marker	–	+	+	–	+	–			+	+	+	+	+	+
Degree adverb	–	–	–	+		+	+		+	+	+	+	+	–

morphemes, each with its own pathway (24). In a given Southern variety, any of those contact morphemes may spell out the *cai* semantics.

- (24) a. *xian* ‘first’: temporal adverb > QF
 b. *zheng* ‘precisely’: adjective > degree / contrastive
 c. *zai* ‘then’: sequential

This differs from the linear sequences proposed by Lin and Li (2014), Li and Wu (2020) and others.²

Hong Kong Cantonese historically runs *zing* > *zai* > *xian* and so inherited

²Lin and Li (2014) give Lianjiang *zheng* (正)—a parallel of Hong Kong *sin*—the pathway adjective > scalar > QF. Unlike *sin*, Lianjiang *zheng* starts as an adjective ‘right’ (i), then develops a scalar use (ii) and a QF use (iii). Lin and Li (2014) render QF *zheng* as Mandarin *daodi*, comparable to English *the hell* (den Dikken and Giannakidou, 2002, Ippolito, 2024, Martin, 2020, Oguro, 2018). Hong Kong *daodi* ‘at all’ can stand in for *sin*_{QF} (iv) and can co-occur with it (v). Both, we suggest, are QUD navigators (Rojas-Esponda, 2014); their relation is left open.

- (i) 牛肉丸 好 正_A 啊
 beef.ball very ZHENG SFP
 ‘The beef balls are really authentic!’

[Lianjiang Cantonese; Lin and Li 2014 ex. 11]

- (ii) 如今正_S [六]_F 點鐘
 now ZHENG six.o’clock
 ‘It is only six o’clock now.’

[Lianjiang Cantonese; Lin and Li 2014 ex. 13]

- (iii) 乜嘢 意思 正_{QF}
 what meaning ZHENG
 ‘What do you mean, then?!’

[Lianjiang Cantonese; Lin and Li 2014 ex. 20]

- (iv) Context: two Annas are friends of both A and B.

A: Anna called.

[Presupposes a unique Anna.]

the core together with the *xian* and *zai* developments. Why the *zheng*-derived contrastive and degree uses did not survive in the modern Cantonese exponent is a question for further research.

4.4 Cross-linguistic parallels

4.4.1 German *denn*

German *denn* resembles Cantonese *sin* in its history.³ Scalar *sin* absorbs *zi* ‘then’; *denn* continues Old High German *thanne* (Abraham, 1991, Wauchope, 1991, Wegener, 2002), modern *dann* ‘then’ (25).

- (25) Der Winter-ausverkauf dauer-t vom 16. bis 27. März.
the winter-clearance last-prs.3SG from.the.DAT.M 16 until 27 March
Dann gilt 50% Rabatt auf alle Winter-artikel.
then apply-prs.3SG 50% discount on all.PL winter-article.PL
‘The winter sale lasts from 16 to 27 March. Then a 50% discount applies

B: dangzan, nei (daodi) gong bingo Anna?
B: wait, 2.SG at.all say which Anna
B: ‘Wait, which Anna did you refer to?’

- (v) Context: same.

B: dangzan, nei (daodi) gong bingo Anna *sin*_{QF}?
B: wait, 2.SG at.all say which Anna SIN
B: ‘Wait, which Anna did you refer to?’

If the two-path analysis of Hong Kong *sin* is right, a strictly linear adjective > scalar > QF sequence for Lianjiang is unlikely to be the whole story.

³As Sze-Wing Tang and Yenan Sun has noted, while *denn* and scalar *sin* are semantically similar, they are syntactically different. While *denn* appears in the antecedent clause, *sin* must appear in the matrix clause. We are not sure whether this syntactic difference will make *denn* differ semantically, nor how syntactically different particles converge to so similar meanings under the compositional semantic approach. We leave this for further analysis.

to all winter items.’ [Siebold 2021 ex. 1]

Unlike *sin*, *denn* does not develop a sequential use (26), but it does develop a conditional use (27).

- (26) A: I just checked my mail. Peter will arrive between 2 p.m. and 4 p.m.
B: Gut, und wenn Peter (#denn) kommt, können wir schwimmen gehen.
B: good and if Peter DENN comes can we swimming go
B: ‘Good, and when Peter comes, we can go swimming.’
[Csipak and Zobel 2014 ex. 5]

- (27) Der Film würde Alex gefallen, wenn er denn käme.
the movie would Alex please if he DENN come.SUBJ
‘Alex would like the movie if he came.’
[Csipak and Zobel 2014 ex. 7]

Relative to a plain conditional, *denn* marks the antecedent as unlikely (28) or counterfactual (29).

- (28) Ich würde mitmachen, wenn ich denn Zeit hätte.
I would join.in if I DENN time had
‘I would join in if I had time. (But I don’t.)’
[Häussler 2015 ex. 4]

- (29) Schüler*innen verbringen ihre Freistunden gerne in der Schülerbücherei
student.FEM spend their free.periods gladly in the school.library
bei der netten Bibliothekarin—sofern es denn eine gäbe.
with the kind librarian if it DENN a be.SUBJ
‘Students like to spend free periods in the school library with the kind
librarian, if there were a school library. (But there isn’t.)’ [Häussler 2015
ex. 7]

Denn is banned when the speaker takes the antecedent to be highly likely (30).

- (30) A: Look, it is sunny!

#B: Stimmt! Wenn es (#denn) sonnig ist, können wir spazieren gehen.
 B: true if it DENN sunny is can we walk go
 B: ‘Right! If it is sunny, we can go for a walk.’

[Csipak and Zobel 2014 ex. 3]

Csipak and Zobel (2014) propose that *denn* asserts that p lies below a contextual threshold of assertability T (31).

$$(31) \quad \llbracket \text{denn} \rrbracket(p) = \lambda w. \text{prob}(w, p) < T \quad [\text{Csipak and Zobel 2014 ex. 24}]$$

In (28) the antecedent is very unlikely, hence below T , and *denn* is defined. *Sin_{Picky}* requires the opposite inequality ($r < p$): the antecedent is stronger, not weaker, than expected. The two directions are not in conflict once likelihood is translated into difficulty: the less likely an antecedent, the more demanding it is. Conditional *denn* and conditional *sin* are therefore close analogues.

Denn also has a question use (32).

(32) Context: A and B know exactly two people called Anna, one in Munich and one in Berlin.

A: Earlier today, Anna called.

B: Welche Anna meinst du denn?

B: which.F Anna mean 2.SG DENN

B: ‘Which Anna do you mean?’

[Theiler 2021 ex. 4]

The addressee has moved to the advanced question *whether Anna called* while the more basic question *which Anna* remains open. The ranking is the same necessary-precondition ranking used for *sin_{QF}* (33).

(33) Context: same as (32).

A: Earlier today, Anna called.

B: nei zi bingo Anna sin?

B: 2.SG refer which Anna SIN

B: ‘Which Anna do you mean, first of all?’

Theiler (2021), Zobel and Csipak (2016) treat both uses of *denn* as signalling a necessary precondition. That analysis does not extend unchanged to *sin*, which has further uses. *Denn* also has a connective use (34) with no Cantonese counterpart; Theiler (2021), Zobel and Csipak (2016) likewise set it aside.

- (34) Es hat geregnet, denn die Straße ist ganz nass.
3.SG.N AUX.3SG PTCP-rain DENN DEF.F street COP.3SG quite wet
‘It was raining, because the street is wet.’

4.4.2 Vietnamese *mới*

Vietnamese *mới* shares the directionality puzzle. Tran and Zimmermann (2021) distinguish an aspectual use (35), an exclusive use (36) and a modal use (37).

- (35) Anh ấy mới_{ASP} (vừa) bước chân ra ngoài.
he PRT just step foot out outside
‘He just stepped outside.’ [Tran and Zimmermann 2021 ex. 1]
- (36) Táo₁ Nam mới_{EXC} ăn *t*₁.
apple Nam PRT eat
‘Nam eats only apples.’ [Tran and Zimmermann 2021 ex. 7]
- (37) Có biên lai mới_{MOD} lấy được sơ-mi.
exist receipt only-then take get shirt
‘You can’t get your shirt until you produce the receipt.’
i. ‘You must produce the receipt in order to get the shirt.’
ii. ‘Only if the receipt exists will you get the shirt.’
[Tran and Zimmermann 2021 ex. 2]

(35) corresponds to adverbial-focus *sin_S* (*aamaam sin* / *sin aamaam*). (36) corresponds to object-focus *sin_S*. Unlike Cantonese in-situ object focus, Vietnamese

exclusive *mới* requires a fronted object (cf. fronted-object *sin_{Picky}*). If the object stays in situ, exclusive *mới* associates with the subject by default. Subject-focus *mới_{EXC}* is always ambiguous with *mới_{ASP}* (38), setting aside the irrelevant adjectival reading ‘new’.

- (38) Sinh viên mới học ngôn ngữ học.
 student PRT learn linguistics
 i. ‘New students learn linguistics.’ [irrelevant]
 ii. ‘The students have just started to learn linguistics.’ [ASP]
 iii. ‘Only students learn linguistics.’ [EXC]
 [Tran and Zimmermann 2021 ex. 5]

(37) covers both the sequential and the conditional uses of *sin*, and is ambiguous between *not-until* and *only-if*.

Tran and Zimmermann (2021) give the entries in (39)–(41).

- (39) $\llbracket \text{mới}_{ASP} \rrbracket = \lambda P_{\langle i, t \rangle} \lambda t. P(t) \wedge \forall t' [t' >_S t] \rightarrow \neg P(t')$, where t is t_{TOP} and $>_S$ is temporal precedence.
 ‘*Mới_{ASP}* asserts that P holds at t and fails at every earlier t' .’
 [Tran and Zimmermann 2021 ex. 22a]
- (40) $\llbracket \text{mới}_{EXC} \rrbracket = \lambda P_{\langle e, it \rangle} \lambda t \lambda x. P(x)(t) \wedge \forall y [y >_S x] \rightarrow \neg P(y)(t)$, where $>_S$ is inclusion on plural individuals.
 ‘*Mới_{EXC}* asserts that P holds of x and fails of every lower y .’
 [Tran and Zimmermann 2021 ex. 22b]
- (41) $\llbracket \text{mới}_{MOD} \rrbracket = \lambda q_{\langle st \rangle} \lambda P_{\langle st, st \rangle} \lambda w. P(q)(w) \wedge \forall Q [Q >_S P \rightarrow \neg Q(q)(w)]$, with $Q >_S P$ iff $Q(q)(w) \rightarrow P(q)(w)$.
 ‘*Mới_{MOD}* asserts that q holds at P and fails at every weaker Q .’
 [Tran and Zimmermann 2021 ex. 40]

Two comments follow from the analysis of *sins*. First, aspectual *mới* is the natural place to look for a scale flip: it corresponds to both *aamaam sin* and *sin aamaam*, so it may itself be ambiguous between a duration reading and a time-point reading, independently of linear order. The focus-direction test used for *sin* does not carry over, because exclusive object focus in Vietnamese always fronts the object. In situ, exclusive object focus is unavailable (42).

- (42) Nam mới ăn táo.
 Nam PRT eat apple
 i. ‘Nam has just eaten the apple(s).’ [aspectual: OK]
 ii. *‘Nam eats only apples.’ [*exclusive object focus]
 iii. #‘Only Nam eats apples.’ [#exclusive subject focus]

A different account of the Vietnamese scale flip is therefore needed; it lies beyond this thesis.

Second, (39) and (40) are exclusive entries, but we doubt that *mới_{ASP}* and *mới_{EXC}* are exclusives. Hole (2017) treats *mới* as scalar and *chỉ* as exclusive; the two co-occur (43).

- (43) Nam chỉ [mỗi thịt bò]_i mới ăn *t_i* thôi.
 Nam only only beef PRT eat *t_i* only
 ‘Only beef does Nam eat.’ [Hole 2017 ex. 1]

It is therefore more likely that *chỉ* carries the exclusive semantics of (40). A further difference: we place $p(w)$ in the presupposition, after Rooth (1992); Tran and Zimmermann (2021) place it in the assertion, because it can be denied (44).

- (44) Không phải là Nam mới đến. Nam chưa đến.
 NEG right C Nam PRT arrive Nam not.yet arrive
 ‘It is not true that Nam just arrived. Nam has not arrived yet.’

For Cantonese *sin_S*, $p(w)$ cannot be denied except by a presupposition-targeting adverbial such as *ganbun* ‘at all’ (45). We therefore keep $p(w)$ presuppositional for Cantonese.

- (45) John m hai aamaam sin lai. keoi #(ganbun) mou lai.
 John NEG cop just.now SIN come 3.SG at.all NEG come
 ‘John did not come just now. He did not come at all.’

On $m\acute{o}i_{MOD}$, compare the *zjau-sin* entry (43) from Chapter 3. Again (41) looks like an exclusive entry, and again p and q are presupposed on our account but asserted on theirs. Their modal scale is disjunctive: $p \vee r$ outranks p because the latter entails the former, so (37) excludes every looser antecedent.

- (46) $(p \vee p' \vee p'') > (p \vee p') > p$
 where p = the receipt exists, p' = the screenshot exists, p'' = some purchase record exists.

Aspectual and exclusive *m\acute{o}i*, by contrast, rank alternatives conjunctively. If the alternatives to (35) were disjointed, every disjunction would collapse to the weakest alternative and no scale would remain. Conjunction preserves rank:

- (47) $(p \wedge p' \wedge p'') > (p \wedge p') > p$
 i.e. one hour > five minutes > three seconds.

The same holds for exclusive *m\acute{o}i*: *Nam only eats apples* excludes the stronger conjunctions *apples and oranges*, *apples and oranges and pears*. Tran and Zimmermann (2021) therefore flip the scale by switching between a conjunctive scale (ASP, EXC) and a disjunctive scale (MOD). We flip the scale by flipping the direction of focus. The two strategies are different answers to the same puzzle.

Vietnamese adfocal *m\acute{o}i* ‘only’ is the counterpart of Cantonese suffixal *dak*. A

bipartite analysis of *only* posits an operator *Op* on the clausal spine and a local head *F* next to the focus (Barbiers, 2014, Cable, 2010, Horvath, 2007, Lee, 2005, Quek and Hirsch, 2017). English bans the two from co-occurring overtly (48).

- (48) *Haki only ate only [a fish]_F. [Sun 2020 ex. 3]

Sun (2020) argues that Vietnamese spells both heads out: *Op* = *chỉ*, *F* = *mỗi* (49)–(50).

- (49) Nam chỉ ăn mỗi [thịt bò]_F
 Nam only_{Adv} eat only_{Adn} beef
 ‘Nam only eats beef.’ [Vietnamese; Sun 2020 ex. 13]
- (50) [mỗi [thịt bò]_F]₁ Nam mới ăn *t*₁.
 only_{Adn} beef Nam PRT eat
 ‘Only beef does Nam eat.’ [Vietnamese; Sun 2020 ex. 21]

Cantonese does the same with *Op* = *zinghai* and *F* = *dak* (51)–(52).

- (51) John zinghai sik dak (jat-faai) [ngaujuk]_F.
 John only_{Adv} eat only_{Adn} one-CL beef
 ‘John only eats beef.’ [Cantonese]
- (52) zinghai [dak [ngaujuk]_F]₁ John sin sik *t*₁
 only_{Adv} only_{Adn} beef John PRT eat
 ‘Only beef does John eat.’ [Cantonese]

As noted in Chapter 3, suffixal *dak* (51) selects a [+Num] DP (Tang, 2000, 2002). Once *dak* is fronted it is no longer a suffix and need not take [+Num]; [+Num] is in fact degraded (53).

- (53) *zinghai [dak [jat]_F-faai ngaujuk]₁ John sin sik *t*₁
 only_{Adv} only_{Adn} one-CL beef John SIN eat
 Int.: ‘Only one piece of beef does John eat.’

Cantonese may therefore divide the labour that Vietnamese leaves to a single pair: in-situ *dak* wants [+Num], fronted *dak* wants [−Num]. A fuller comparison is left open.

4.4.3 *Only* and *until*

Only and *not-until* are near-reflections of each other. The parallelism is cross-linguistic. Giannakidou (2002) distinguishes two Greek *untils*: PPI *mexri* and NPI *para monon*. The latter is morphologically *para* ‘but’ plus *monon* ‘only’ (54).

- (54) I prigipisa *(dhen) eftase para monon ta mesanixta.
 the princess not arrived.PERF.3SG but only the midnight
 ‘The princess did not arrive until midnight.’
 = ‘It was only at midnight that the princess arrived.’

[Greek; Giannakidou 2002 ex. 36]

French *ne...que* pairs negation with *que* ‘until’ (55); Italian *non...che* does the same, with the concord *non* preceding *che* (56).⁴

⁴The *non* after *che* is expletive. Expletive negation after *until* is also found in Slovenian (i) and Croatian (ii) (Abels, 2005, Gradimondo, 2024, Portner and Zanuttini, 2000, Zovko and Ilc, 2017).

- (i) Pes bo cvilil, vse dokler ga {*nehaš/ ne nehaš} tepsti.
 dog whine-FUT until it-ACC *stop-2SG.PRES/ not stop-2SG.PRES beat
 ‘The dog will whine until you stop beating it.’ [Slovenian; Zovko and Ilc 2017 ex. 10a]
- (ii) Skrivat ću se dok (ne) svane.
 hide AUX.FUT.1SG REFL until (not) dawn.PRES.3SG
 ‘I’ll hide until it dawns.’ [Croatian; Zovko and Ilc 2017 ex. 10b]

(55) Jean ne partira qu'à cinq heures.
 John NEG leave-FUT.3SG QUE at five-hour.PL
 'John will not leave until five.' [French; Declerck 1995, p. 59]

(56) et non è da fidare in loro infin che non son conosciuti.
 and NEG is to to.trust in them until that EXPL be.3.PL known
 'And do not trust them until they have been well known.'
 [Italian; Faleri 2009, p. 199]

A broader typology is de Swart et al. (2022). Outside Europe, Olguin Martinez (2023) finds *until* developing from *only* in several Australian languages, for example Biniñ Gun-Wok *djal* (57), and treats the development as a Pama-Nyungan areal feature. Bilinarra =rni 'until' likewise continues adverbial 'only' (58).

(57) gabarri-djal-noihme-noihme dja ga-rrung-bebme.
 3-cvb.only-iter.sleep.with-RDP and 3SG-sun-appear
 'They just keep going until the sun comes up.'
 [Biniñ Gun-Wok; Evans 2003, p. 657]

(58) aru-nggu dirl ba-ni marluga ngarlaga-ngga
 child-ERG hit.head hit-PST old.man head-LOC
 gungulu-g-ba=rni
 bleed-fact-EP=CONJ.ONLY
 'A child hit the old man on the head until he bled.'
 [Bilinarra; Meakins and Nordlinger 2014, p. 386]

The cross-linguistic record, then, is that *only*, *until* and negation form a tight cluster—exactly the cluster that scalar *sin* exploits.

Chapter 5

Loose ends

The uses of *sin* and the scales they associate with are summarized in Table 5.1.

Table 5.1: Varieties of *sin* and associated scales

Variety of <i>sin</i>	Associated scale
Temporal <i>sin_T</i>	Time-point scale $<_{TP}$
<i>So-few sin_{Few}</i>	Number scale $<_N$ (with overt Num ⁰)
<i>So-late sin_{Late}</i>	Time-point scale $<_{TP}$
<i>So-picky sin_{Picky}</i>	Condition scale $<_C$
Adverbial-focus scalar <i>sin</i>	$<_{TP}$ or $<_N$
Question-focus <i>sin_{QF}</i>	Necessary-precondition scale $<_Q$

Two issues remain open.

5.1 When is *sin* obligatory?

In Mandarin, *cai* is optional under rightward focus: it only adds a LONG evaluation, as in the *so-few* use (1).

- (1) John (cai) chi-le [san]_F-ge pinggwo.
 John CAI eat-PRV three-CL apple
 ‘John just ate three apples.’ [Mandarin]

The same holds for Cantonese *sin_{Few}* in the rightward case (2).

- (2) John (sin) sik-dak [saam]_F-go pinggwo.
 John SIN eat-ONLY three-CL apple
 ‘John just ate three apples.’ [Cantonese]

Under *leftward* focus, however, both particles become obligatory: the *so-picky* uses (3)–(4) and the *so-late* uses (5)–(6).

- (3) Mary qu paidui, John *(cai) qu paidui.
 Mary go party John CAI go party
 ‘Only if Mary goes to the party will John go.’ [Mandarin]

- (4) Mary heoi party, John *(sin) heoi party.
 Mary go party John SIN go party
 ‘Only if Mary goes to the party will John go.’ [Cantonese]

- (5) yanchanghui jieshu-le, John *(cai) lai.
 concert finish-PRV John CAI come
 ‘It was not until the concert finished that John came.’ [Mandarin]

- (6) jincoengwui jyun-zo, John *(sin) lai.
 concert finish-PRV John SIN come
 ‘It was not until the concert finished that John came.’ [Cantonese]

A promising account is Maximize Presupposition (Heim, 1991). Sun and Ng (Forthcoming), building on Liu’s (2021b) MP! treatment of Mandarin *mei-dou* and on degree analyses of English *even* (Greenberg, 2016, 2018, Zhang, 2022), argue that *cai_{Few}* answers a quantity question (‘How many x did P ?’), whereas *cai_{Late}* and *cai_{Picky}* answer a quality question (‘How G_{QUD} is x_{QUD} ?’). Only the quality question coerces a degree-exclusivity presupposition (every alternative is

excluded), and so *cai* is obligatory there. MP! is compatible with the present analysis and would explain the asymmetry. The three configurations, which extend immediately to *cai*, are repeated in Figures 5.1–5.3.

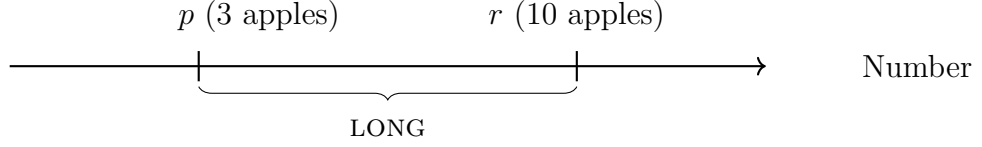


Figure 5.1: The *so-few* use

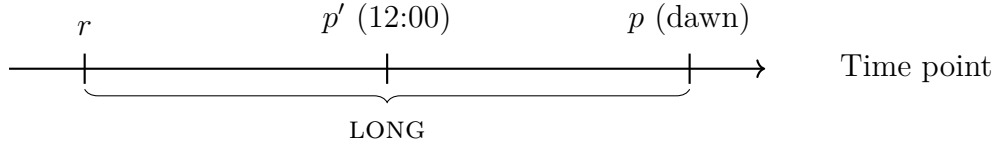


Figure 5.2: The sequential use

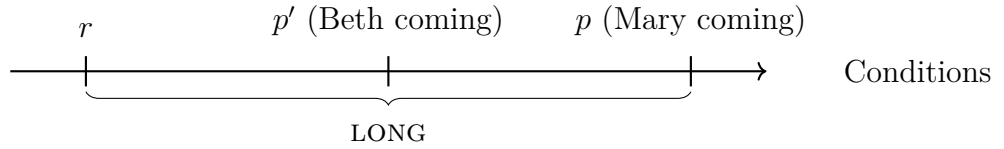


Figure 5.3: The picky use

For *sin_{Late}* and *sin_{Picky}*, a competitor p' sits between p and r . That competitor must be taken into account when the distance from r to p is measured. Because p' is closer to r than p is, the r – p gap is coerced as LONG, and the marker is obligatory. For *sin_{Few}* nothing sits between p and r , so no relative distance forces a LONG evaluation: p and r may go unevaluated, or even be evaluated as SHORT. Hence *sin_{Few}* is optional. This is only a sketch; a full MP! implementation is left

for future work.¹

¹Fronted-object *sin_{Picky}* looks like a counterexample. In Mandarin, dropping *cai* is bad at first glance (i), but the sentence improves with an experiential suffix or a modal (ii) (Sun, 2021, ex. 14).

- (i) ??zhiyou [niurou]_F Yuehan chi.
 only beef John eat
 Int.: ‘Only [beef]_F did John eat.’ [Mandarin]
- (ii) zhiyou [niurou]_F Yuehan {chi-guo/ xihuan chi}.
 only beef John eat-EXP like eat
 ‘Only beef has John eaten before.’ / ‘Only beef does John like to eat.’ [Mandarin]

The same pattern holds in Cantonese. If genuine, it would allow the scalar marker to be missing even under leftward focus. We suspect that (ii) is VP topicalization, with *zhiyou* as the predicate, underlyingly as in (iii).

- (iii) Yuehan {??chi/ ^{OK}chi-guo/ ^{OK}xihuan chi} de zhiyou niurou.
 John eat / eat-EXP / like eat NMLZ only beef
 With *-le*, int.: ‘What John has eaten is only beef.’
 With *-guo*: ‘What John has eaten is only beef.’
 With *xihuan*: ‘What John likes to eat is only beef.’ [Mandarin]

After topicalization the nominalizer *de* is silent. The grammaticality facts line up: (iii), like (i), bans simple perfective *-le* and allows experiential *-guo* and modal *xihuan*. Crucially, *cai* cannot be added to (iii), as in (iv).

- (iv) *Yuehan cai xihuan chi de zhiyou pingguo.
 John CAI like eat NMLZ only apple
 Int.: ‘What John likes to eat is only apples. (So few.)’

On this view (ii) is a different construction that happens to look similar, and is not a counterexample to the leftward-obligatory generalization.

5.2 The emphatic use

Lai (1999) identifies an emphatic *cai* that highlights the polarity of the sentence. The use may be positive or negative; both require a preceding utterance to deny. The positive cases are (7) and (8). Incompatibility with *zhijou/zijau* shows that they are not *sin_{Picky}*.

- (7) A: Mary is the smartest.
 B: (#zhijou) Beth_F cai shi zui congming!
 B: only.have Beth CAI COP most smart
 B: ‘BETH is the smartest!’ [Mandarin]
- (8) A: Mary is the smartest.
 B: (#zijau) Beth_F sin hai zeoi cungming!
 B: only.have Beth SIN COP most smart
 B: ‘BETH is the smartest!’ [Cantonese]

The negative case co-occurs with focused negation, as in (9) and (10).

- (9) A: It’s cold outside!
 B: ceotmin sin [m]_F dung aa3!
 B: outside SIN NEG cold SFP
 B: ‘It’s NOT cold outside!’ [Cantonese]
- (10) A: It’s cold outside!
 B: waimian cai [bu]_F leng ne!
 B: outside CAI NEG cold SFP
 B: ‘It’s NOT cold outside!’ [Mandarin]

We leave the positive emphatic use open. For the negative use, a preliminary hypothesis is that it is a *FALSUM* operator; see Appendix B.

5.3 Conclusion

This thesis has argued for two independent developments from temporal *sin_T*. One path type-shifts the sequencing semantics of *sin_T* from propositions to questions and yields *sin_{QF}*, a QUD navigator ordered by necessary preconditionhood. The other path absorbs the scalar force of historical *zi* and yields *sin_S*, a single operator that contributes a LONG presupposition and combines with exclusive particles by Predicate Modification. Once the exclusive component is factored out, the directionality puzzle is no longer a fact about two lexical entries for *sin*. It is a fact about which exclusive *sin* combines with, and about whether the induced scale measures ‘how little’ or ‘how late / how strong’. The same decomposition extends, with local adjustments, to Cantonese *zau*, Mandarin *cai* and a range of Southern Sinitic and extra-Sinitic particles. What remains—the MP! account of obligatoriness, the positive emphatic use, and a fully unified treatment of *zau/jiu*—are natural next steps rather than obstacles to the analysis.

Appendix A

Index of glossary terms

Adv Adverbial	NEG Negation
AI At-issue content; Actuality inference	PM Predicate Modification
ASP Aspectual	PRV Perfective
BEI Beyond-expectation inference	QF Question-focus
CG Common Ground	RC Relative Clause
CL Classifier	SFP Sentence-final particle
Cond Conditional	<i>sin_T</i> Temporal <i>sin</i>
CP Complementizer Phrase	<i>sin_{QF}</i> Question-focus <i>sin</i>
EXC Exclusive	<i>sin_S</i> Scalar <i>sin</i>
FALSUM FALSUM	<i>sin_{Few}</i> <i>So-few</i> scalar <i>sin</i>
MOD Modal	<i>sin_{Late}</i> <i>So-late</i> scalar <i>sin</i>
	<i>sin_{Picky}</i> <i>So-picky</i> scalar <i>sin</i>

Appendix B

Negative emphatic *sin* as falsum

An example of negative emphatic *sin* is (1).

- (1) A: It is cold outside!
B: ceotmin sin [m]_F dung aa3!
B: outside SIN NEG cold SFP
B: ‘It is NOT cold outside!’

Romero and Han’s (2004) entry for FALSUM is (2).

- (2) $\llbracket \text{FALSUM} \rrbracket = \lambda p : \forall w' \in \text{Epi}_{\text{Origo}_c}(w) [\forall w'' \in \text{Conv}_{\text{Origo}}(w') [p \notin \text{CG}]] . \neg p$
‘*Origo_c* is sure that, in all worlds compatible with their conversational goals, *p* is not in the Common Ground.’

FALSUM is therefore infelicitous out of the blue. The same holds for negative emphatic *sin*, as in (3).

- (3) A bumped into B on the street and said:
#ceotmin sin [m]_F dung aa3!
#outside SIN NEG cold SFP
A int.: ‘It is NOT cold outside!’

A FALSUM alternative competes with plain negation. By the Principle of Economy (Romero and Han, 2004), FALSUM is blocked unless the speaker attributes to the addressee a bias toward p .

- (4) Do not use a meta-conversational move unless necessary.

FALSUM therefore cannot occur out of the blue: the addressee must already be committed to the opposite. English has no dedicated FALSUM morphology; the effect is often achieved syntactically (for example by neg-raising). Frana and Rawlins (2016, 2019) argue that Italian *mica* is FALSUM morphology. *Mica* occurs alone (the autonomous use, $mica_A$) or together with negation (the discontinuous use, $mica_D$); both yield a FALSUM reading. $Mica_A$ both negates and marks FALSUM, so it is not truth-conditionally vacuous (5).

$$(5) \quad \llbracket mica_A \rrbracket = \lambda p : \forall w' \in Epi_{Origoc}(w) [\forall w'' \in Conv_{Origo}(w') [p \notin CG]]. \neg p$$

$Mica_D$ co-occurs with negation and is therefore truth-conditionally vacuous: negation does the negating. Negative emphatic *sin* likewise requires a co-occurring negation, so it too is truth-conditionally vacuous, on the model of (6).

$$(6) \quad \llbracket mica_D \rrbracket = \lambda \phi : \forall w' \in Epi_{Origoc}(w) [\forall w'' \in Conv_{Origo}(w') [\phi \in CG]]. \phi$$

, where ϕ is a negated proposition.

The problem for a unified account is that (6) makes no reference to a scale. The intuition, however, is scalar. Negative emphatic *sin* does not merely say that the prejacent is false; it says that the prejacent is **a far cry from the truth**, which is the LONG evaluation of scalar *sin*. We can therefore set up a two-point scale of truth on which TRUE outranks FALSE. The particle marks the distance between those two alternatives as long: what the addressee believed was a far cry from

the truth, as in Figure B.1.

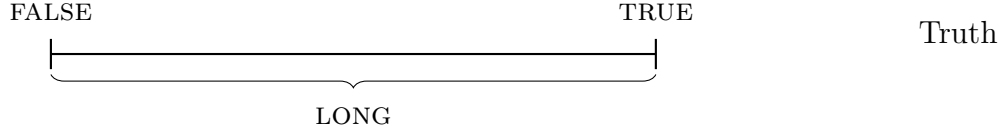


Figure B.1: FALSUM *sin*

A two-point scale is still a scale. Compositionally we assume the LF in (7).

$$(7) \quad \text{LF for negative emphatic } \textit{sin}: [\text{sin} [\text{m} [p]]]$$

Negation has its standard entry.

$$(8) \quad \llbracket \text{m} \rrbracket = \lambda p \lambda w. \neg p(w)$$

Negative emphatic *sin* then takes the already negated proposition by Functional Application, as in (9).¹

$$(9) \quad \llbracket \text{sin}_{\text{Falsum}} \rrbracket(C_i) = \lambda q \lambda w : \exists r [q <_T r \wedge \text{LONG}(q, r)]. q(w)$$

, where $<_T$ is a scale of truth and q is a negated proposition.

‘ $\text{Sin}_{\text{Falsum}}$ presupposes that q is a far cry from r (the truth).’

Saying that $\neg p$ is a far cry from the truth amounts to saying that every world compatible with the speaker’s conversational goals is a $\neg p$ -world. Plain negation does not presuppose that universal quantification, so p and $\neg p$ remain in dispute—and p is then not a far cry from the truth. (9) can therefore be rewritten as (10), in line with Romero and Han (2004).

¹PM is unavailable here. In the PM cases both the scalar particle and the exclusive manipulate p . Here *sin* does not manipulate p ; it manipulates $\neg p$, and so must take the negated proposition as argument. A full comparison of the PM and FA regimes for *sin* is left open.

- (10) $\llbracket \text{sin}_{\text{Falsum}} \rrbracket = \lambda q \lambda w : \forall w' \in \text{Epi}_{\text{Origo}_c}(w) [\forall w'' \in \text{Conv}_{\text{Origo}}(w') [q \in CG]]. q(w)$
, where q is a negated proposition.

Before closing we consider *FALSUM sin* in questions. Frana and Rawlins (2016, 2019) argue that Italian *mica* undergoes bias reversal (Garrett, 2001) in questions because of origo shift (Bhadra, 2017, 2020, Speas and Tenny, 2003, Wang, 2025a, Wang and Cheung, 2025). The evaluator anchors to the speaker in assertions and to the addressee in questions. *Mica* is subject to that shift only if Q scopes over it, as in (11).

- (11) a. LF of a *mica* assertion: $[mica [p]]$
b. LF of a *mica* question: $[Q [mica [p]]]$

Origo shift would challenge the present account, whose central claim is that scalar *sin* encodes the *speaker's* evaluation. We suggest that the shift does not apply to Cantonese *sin*: the origo remains the speaker even in questions. As already noted, *FALSUM sin* is incompatible with a *wh*-constituent (12), with A-not-A reduplication (13) and with alternative questions (14).

- (12) *bingo sin $[m]_F$ hai lousi?
who SIN NEG COP teacher
Int.: ‘Who is not a teacher? (What you said was a far cry from the truth!)’
- (13) *John sin $[m]_F$ hai m hai heoi-zo jatbun?
John SIN NEG COP NEG COP go-PRV Japan
Int.: ‘Did John not go to Japan? (What you said was a far cry from the truth!)’
- (14) *John sin $[m]_F$ jiu caa dinghai gaafe?
John SIN NEG want tea or coffee
Int.: ‘Does John not want tea or coffee? (What you said was a far cry from the truth!)’

The only way to put FALSUM *sin* in a question is with a confirmational such as *o3ho2* (Lam, 2014, Law et al., 2018, 2024, Ng, 2025, Tang, 2020). Compare (15) with (16).

- (15) Context: Mary loved John; John no longer loves Mary and has cheated.
Mary cannot believe it and asks John.

nei sin mou ceotgwai o3ho2?
2.SG SIN NEG cheat SFP
‘You did NOT cheat on me, right?’

- (16) Context: John loved Mary; Mary no longer loves John. John cheated but keeps insisting that he did not. Mary asks John.

#nei sin mou ceotgwai o3ho2?
2.SG SIN NEG cheat SFP
‘You did NOT cheat on me, right?’

In (15) Mary is the origo: it is Mary who wants it settled that John did not cheat. In (16) the origo would have to be John, the only person who knows whether he cheated. Because (15) is felicitous and (16) is not, *sin* still anchors to the speaker in questions.^{2 3}

²In (i) *sin* is licensed even though Mary no longer loves John, because Mary is quoting John, and the sarcasm is parasitic on that quote. The origo remains the original speaker of *sin*, namely John.

- (i) Context: John loved Mary; Mary no longer loves John. John cheated but keeps insisting that he did not.

John: ngo sin mou ceotgwai aa3!
John: 1.SG SIN NEG cheat SFP
John: ‘I did NOT cheat!’

Mary: nei sin mou ceotgwai o3ho2?
Mary: 2.SG SIN NEG cheat SFP
Mary: ‘You did not cheat, huh?’

³Formally, the pieces in (i) combine by FA as in (ii). Two remarks on *o3ho2*. First, it is

Negative emphatic *sin* is therefore incompatible with interrogative syntax proper: the only host is a confirmational, whose prejacent remains declarative. Origo shift does not apply; the origo stays with the speaker. If both the present analysis and [Frana and Rawlins \(2016, 2019\)](#) are on the right track, natural languages have at least two FALSUM operators: one that always anchors to the speaker (Cantonese *sin*), and one that undergoes origo shift (Italian *mica*).

a positively biased question; bias is marked in **bold** ([Farkas and Roelofsen, 2017](#)). Second, the Hamblin negative copy ([Hamblin, 1973](#)) is a metalinguistic negation: it flips both the assertion and the presupposition. Ordinary negation would assert that the addressee cheated while presupposing that, for all of the speaker’s conversational goals, the addressee did not—a contradiction.

- (i) a. $\llbracket \text{nei mou ceotgwai} \rrbracket = \lambda w. \neg \text{CHEAT}(j, w)$
- b. $\llbracket \text{sin} \rrbracket = \lambda q \lambda w : \exists r [q < r \wedge \text{LONG}(q, r)]. q(w)$
- c. $\llbracket \text{o3ho2} \rrbracket = \lambda p. \{ \lambda \mathbf{w}. \mathbf{p}(\mathbf{w}), \lambda w. \neg p(w) \}$

- (ii) LF: $[\text{o3ho2} [\text{sin } [p]]]$

The denotation of [\(15\)](#) is then [\(iii\)](#).

- (iii) $\{ \lambda w : \exists r [\text{LONG}(\neg \text{CHEAT}(j, w), r)]. \neg \text{CHEAT}(j, w),$
 $\lambda w : \neg \exists r [\text{LONG}(\neg \text{CHEAT}(j, w), r)]. \text{CHEAT}(j, w) \}$

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